UNIVERSITY WORKBOOK SERIES IN CHEMICAL SCIENCES

STRUCTURE OF MATTER

Complete Problem Workbook with Tablet Workspaces & Full Solutions

Progressive Exercises from Scratch to Mastery

Complete Step-by-Step Solutions After Every Single Exercise

Featuring 75 meticulously structured exercises across solution concentrations, normality, redox titrations, isotope abundances, mass defects, the Aston curve, radioactive decay kinetics, and radiometric dating.

Equipped with dedicated **5 mm graphic tablet workspaces** formatted for direct digital stylus problem-solving in Xournal++, OneNote, or PDF annotation software.

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Department of Sciences · First Edition

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How to Use This Workbook

This problem workbook is engineered specifically for active, hands-on learning with a **graphic tablet or digital stylus** on PC.

Key Pedagogical Features

1. **Progressive Difficulty Rating:** Every exercise is categorized by difficulty:

   ★ **Foundational:** Direct application of fundamental definitions, units, and single-step formulas.

   ★★ **Intermediate:** Multi-step problems requiring stoichiometric conversions, systems of linear equations, or differential decay integration.

   ★★★ **Advanced / Exam Level:** Comprehensive synthetic problems, commercial reagent dilutions, redox titrations with varied valence factors, and radiometric dating with error propagation.

2. **Dedicated 5 mm Tablet Grid Workspaces:** Each exercise is immediately followed by an authentic 5 mm Cartesian grid box calibrated for standard tablet stylus resolution. Open this PDF in your favorite annotation application (such as **Xournal++**, **Drawboard**, or **OneNote**) to solve problems by hand.

3. **Immediate, Complete Solutions:** Following each workspace, a comprehensive step-by-step solution provides every algebraic manipulation, unit conversion, and final numerical result.

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Chapter 1

Fundamental Concepts and Solution Chemistry

Essential Chapter Formulas & Rules · Core Definitions, Concentration Scales & Dilution Laws

1. **Amount of Substance (Mole) & Avogadro's Number:**

$$n = \frac{m}{M} = \frac{N}{N_A} = \frac{V_{\text{gas}}}{V_0} \quad \text{where } N_A = 6.022\,141 \times 10^{23}\ \text{mol}^{-1},\ V_0 = 22.414\ \text{L} \cdot \text{mol}^{-1}\ (\text{STP}).$$

2. **Molarity (C_M or [X]) & Mass Concentration (C_m):**

$$C_M = \frac{n_{\text{solute}}}{V_{\text{sol}}} \quad [\text{mol} \cdot \text{L}^{-1} \equiv \text{M}], \qquad C_m = \frac{m_{\text{solute}}}{V_{\text{sol}}} \quad [\text{g} \cdot \text{L}^{-1}], \qquad C_m = C_M \cdot M.$$

3. **Molality (m or b) & Commercial Reagent Relation:**

$$m = \frac{n_{\text{solute}}}{m_{\text{solvent}}\ (\text{kg})} \quad [\text{mol} \cdot \text{kg}^{-1}], \qquad C_M = \frac{10 \cdot P\% \cdot d}{M} \quad (d = \rho_{\text{sol}}/\rho_{\text{water}}).$$

4. **Normality (N) & Equivalent Mass (E_q):**

$$N = Z \cdot C_M = \frac{m_{\text{solute}}}{E_q \cdot V_{\text{sol}}}, \qquad E_q = \frac{M}{Z}.$$

- *Acids/Bases:* Z = number of exchangeable H^+ or OH^-.
- *Salts/Precipitates:* Z = total positive charge of metal cations = $n_{\text{cation}} \times z^+$.
- *Redox:* Z = number of electrons transferred per mole of reagent.

5. **Dilution Law & Titration Equivalence Principle:**

$$n_{\text{initial}} = n_{\text{final}} \implies C_1V_1 = C_2V_2, \qquad \text{Equivalence:} \quad N_1V_1 = N_2V_2.$$

6. **Mixing Non-reacting Solutions of the Same Solute:**

$$C_{\text{final}} = \frac{n_1 + n_2 + \ldots}{V_1 + V_2 + \ldots} = \frac{C_1V_1 + C_2V_2 + \ldots}{V_{\text{total}}}.$$

Exercise 1.1 · States of Matter and Phase Transition Enthalpy ★ Foundational

Classify the following physical transformations according to their thermodynamic nature (endothermic or exothermic), state the phase transition name, and describe the molecular-level change in kinetic and potential energy:

1. Formation of frost on a cold winter morning directly from atmospheric water vapor.

2. The disappearance of dry ice ($CO_2(s)$) placed on a laboratory bench at room temperature without liquid formation.
3. The condensation of steam on a chilled condenser coil during a distillation experiment.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.1

1. Frost Formation (Water vapor $\longrightarrow$ Solid ice):

- **Transition Name: Deposition** (or desublimation / solid condensation).
- **Thermodynamic Nature: Exothermic** ($\Delta H_{\text{dep}} < 0$). Heat is released into the surroundings.
- **Microscopic Description:** High-energy, disordered gaseous water molecules lose kinetic energy upon contact with the cold surface. Intermolecular hydrogen bonds form, locking water molecules into an ordered crystalline lattice with minimal potential energy.

2. Disappearance of Dry Ice ($CO_2(s) \longrightarrow CO_2(g)$):

- **Transition Name: Sublimation**.
- **Thermodynamic Nature: Endothermic** ($\Delta H_{\text{sub}} > 0$). Heat is absorbed from the room.
- **Microscopic Description:** Thermal energy from the room overcomes the weak van der Waals dispersion forces holding non-polar CO_2 molecules in the solid lattice, increasing both intermolecular separation (potential energy) and molecular velocity (kinetic energy).

3. Condensation of Steam ($H_2O(g) \longrightarrow H_2O(l)$):

- **Transition Name: Condensation** (or liquefaction).
- **Thermodynamic Nature: Exothermic** ($\Delta H_{\text{cond}} = -\Delta H_{\text{vap}} < 0$).
- **Microscopic Description:** Heat is transferred to the condenser coolant, decreasing molecular velocities. Attractive dipole-dipole and hydrogen-bonding interactions pull vapor molecules together into a dense, fluid condensed liquid phase.

Exercise 1.2 · The Mole Concept and Avogadro Counting ★ Foundational

A pure sample of metallic iron (Fe) has a mass of $m = 16.755$ g.

1. Calculate the amount of substance n of iron in the sample ($M(\text{Fe}) = 55.845\ \text{g} \cdot \text{mol}^{-1}$).
2. Compute the absolute number of iron atoms N_{atoms} contained in this sample using $N_A = 6.022\,141 \times 10^{23}\ \text{mol}^{-1}$.

3. What mass of elemental sulfur (S, $M = 32.065\ \text{g} \cdot \text{mol}^{-1}$) contains exactly the same number of atoms as this iron sample?

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.2

1. Amount of Substance of Iron (n_{Fe}):

$$n_{\text{Fe}} = \frac{m_{\text{Fe}}}{M(\text{Fe})} = \frac{16.755\ \text{g}}{55.845\ \text{g} \cdot \text{mol}^{-1}} = \mathbf{0.3000}\ \text{mol}.$$

2. Absolute Number of Iron Atoms (N_{atoms}):

$$N_{\text{atoms}} = n_{\text{Fe}} \cdot N_A = 0.3000\ \text{mol} \times 6.022\,141 \times 10^{23}\ \text{mol}^{-1} = \mathbf{1.8066 \times 10^{23}}\ \text{atoms}.$$

3. Equivalent Mass of Sulfur: To contain identical numbers of atoms, the amounts of substance must be identical: $n_{\text{S}} = n_{\text{Fe}} = 0.3000$ mol.

$$m_{\text{S}} = n_{\text{S}} \cdot M(\text{S}) = 0.3000\ \text{mol} \times 32.065\ \text{g} \cdot \text{mol}^{-1} = \mathbf{9.6195}\ \text{g}.$$

Exercise 1.3 · Molar Mass of Hydrated and Polyatomic Crystals ★ Foundational

Calculate the molar mass of the following chemical compounds to two decimal places:

1. Anhydrous calcium phosphate: $Ca_3(PO_4)_2$.
2. Copper(II) sulfate pentahydrate: $CuSO_4 \cdot 5\,H_2O$.
3. Potassium permanganate: $KMnO_4$.
4. Ammonium iron(II) sulfate hexahydrate (Mohr's Salt): $(NH_4)_2Fe(SO_4)_2 \cdot 6\,H_2O$.

Standard atomic masses ($\text{g} \cdot \text{mol}^{-1}$): H = 1.008, N = 14.007, O = 15.999, P = 30.974, S = 32.065, K = 39.098, Ca = 40.078, Mn = 54.938, Fe = 55.845, Cu = 63.546.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.3

1. Calcium phosphate $Ca_3(PO_4)_2$:

$$\begin{aligned} M &= 3 \cdot M(\mathrm{Ca}) + 2 \cdot M(\mathrm{P}) + 8 \cdot M(\mathrm{O}) \\ &= 3(40.078) + 2(30.974) + 8(15.999) = 120.234 + 61.948 + 127.992 = \mathbf{310.17}\ \mathrm{g \cdot mol^{-1}}. \end{aligned}$$

2. Copper(II) sulfate pentahydrate $CuSO_4 \cdot 5\,H_2O$:

$$\begin{aligned} M(\mathrm{CuSO_4}) &= 63.546 + 32.065 + 4(15.999) = 159.607\ \mathrm{g \cdot mol^{-1}} \\ M(5\,\mathrm{H_2O}) &= 5[2(1.008) + 15.999] = 5(18.015) = 90.075\ \mathrm{g \cdot mol^{-1}} \\ M_{\text{total}} &= 159.607 + 90.075 = \mathbf{249.68}\ \mathrm{g \cdot mol^{-1}}. \end{aligned}$$

3. Potassium permanganate $KMnO_4$:

$$M = 39.098 + 54.938 + 4(15.999) = 39.098 + 54.938 + 63.996 = \mathbf{158.03}\ \mathrm{g \cdot mol^{-1}}.$$

4. Mohr's salt $(NH_4)_2Fe(SO_4)_2 \cdot 6\,H_2O$:

$$\begin{aligned} M &= 2[14.007 + 4(1.008)] + 55.845 + 2[32.065 + 4(15.999)] + 6(18.015) \\ &= 2(18.039) + 55.845 + 2(96.061) + 108.090 = 36.078 + 55.845 + 192.122 + 108.090 \\ &= \mathbf{392.14}\ \mathrm{g \cdot mol^{-1}}. \end{aligned}$$

Exercise 1.4 · Molar Volume and Gas Stoichiometry at STP and RTP ★ Foundational

A sealed rigid gas cylinder contains $V = 5.600$ L of carbon dioxide gas (CO_2) under standard conditions of temperature and pressure (STP : $T_0 = 273.15$ K, $P_0 = 1.000$ atm, where $V_0 = 22.414\ \mathrm{L \cdot mol^{-1}}$).

1. Determine the amount of substance n of CO_2 in the cylinder.
2. Compute the mass of CO_2 gas present ($M(\mathrm{CO_2}) = 44.01\ \mathrm{g \cdot mol^{-1}}$).
3. What volume would this exact same mass of gas occupy at room temperature and pressure (RTP : $T_1 = 298.15$ K, $P_1 = 1.000$ atm)?

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.4

1. Amount of Substance at STP:

$$n = \frac{V}{V_0} = \frac{5.600\ \text{L}}{22.414\ \text{L} \cdot \text{mol}^{-1}} = \mathbf{0.2498}\ \text{mol} \approx \mathbf{0.250}\ \text{mol}.$$

2. Mass of CO_2 Gas:

$$m = n \cdot M(\text{CO}_2) = 0.24983\ \text{mol} \times 44.01\ \text{g} \cdot \text{mol}^{-1} = \mathbf{10.995}\ \text{g} \approx \mathbf{11.00}\ \text{g}.$$

3. Gas Volume at RTP (298.15 K, 1.000 atm): Using the ideal gas law $PV = nRT$ or Charles's law ($V_1/T_1 = V_0/T_0$ at constant pressure):

$$V_{\text{RTP}} = V_{\text{STP}} \cdot \frac{T_{\text{RTP}}}{T_{\text{STP}}} = 5.600\ \text{L} \times \frac{298.15\ \text{K}}{273.15\ \text{K}} = \mathbf{6.112}\ \text{L}.$$

Alternatively, using the RTP molar volume $V_{m,\text{RTP}} = \frac{RT}{P} = \frac{0.08206 \times 298.15}{1.000} = 24.466\ \text{L} \cdot \text{mol}^{-1}$:

$$V = n \cdot V_{m,\text{RTP}} = 0.24983\ \text{mol} \times 24.466\ \text{L} \cdot \text{mol}^{-1} = \mathbf{6.112}\ \text{L}.$$

Exercise 1.5 · Molarity and Mass Concentration of Glucose ★ Foundational

An intravenous infusion bag contains $m = 25.00$ g of pharmaceutical anhydrous glucose ($C_6H_{12}O_6$) dissolved in distilled water to make a total solution volume of $V = 500.0$ mL.

1. Compute the molar mass M of D-glucose ($M(\text{C}) = 12.011$, $M(\text{H}) = 1.008$, $M(\text{O}) = 15.999\ \text{g} \cdot \text{mol}^{-1}$).
2. Determine the mass concentration C_m of glucose in $\text{g} \cdot \text{L}^{-1}$.
3. Determine the molarity C_M of glucose in $\text{mol} \cdot \text{L}^{-1}$ (M).
4. Verify explicitly the fundamental identity $C_m = C_M \cdot M$.

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Complete Step-by-Step Solution 1.5

1. Molar Mass of Glucose:

$$M(\mathrm{C_6H_{12}O_6}) = 6(12.011)+12(1.008)+6(15.999) = 72.066+12.096+95.994 = \mathbf{180.16}\ \mathrm{g\cdot mol^{-1}}.$$

2. Mass Concentration C_m:

$$C_m = \frac{m_{\text{solute}}}{V_{\text{solution}}} = \frac{25.00\ \mathrm{g}}{0.5000\ \mathrm{L}} = \mathbf{50.00}\ \mathrm{g\cdot L^{-1}}.$$

3. Molarity C_M:

$$n = \frac{m}{M} = \frac{25.00\ \mathrm{g}}{180.16\ \mathrm{g\cdot mol^{-1}}} = 0.13877\ \mathrm{mol}.$$

$$C_M = \frac{n}{V} = \frac{0.13877\ \mathrm{mol}}{0.5000\ \mathrm{L}} = \mathbf{0.2775}\ \mathrm{mol\cdot L^{-1}} \equiv \mathbf{0.2775}\ \mathrm{M}.$$

4. Verification of $C_m = C_M \cdot M$:

$$C_M \cdot M = 0.27754\ \mathrm{mol\cdot L^{-1}} \times 180.16\ \mathrm{g\cdot mol^{-1}} = \mathbf{50.00}\ \mathrm{g\cdot L^{-1}} \equiv C_m. \quad \text{(Verified)}$$

Exercise 1.6 · Molarity vs. Molality: Temperature Dependence ★ Foundational

A solution is prepared by dissolving $m_s = 18.02$ g of urea ($\mathrm{CO(NH_2)_2}$, $M = 60.06\ \mathrm{g\cdot mol^{-1}}$) in $m_w = 250.0$ g of pure water. The final volume of the solution at 20°C is measured to be $V = 262.5$ mL.

1. Calculate the molality m of the urea solution in $\mathrm{mol\cdot kg^{-1}}$.
2. Calculate the molarity C_M of the urea solution at 20°C in $\mathrm{mol\cdot L^{-1}}$.
3. If the solution is heated to 80°C, the liquid expands thermally to a volume of 268.0 mL. Explain whether molality or molarity changes, and calculate the new values.

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Complete Step-by-Step Solution 1.6

1. Molality m:

$$n_{\text{urea}} = \frac{18.02\ \text{g}}{60.06\ \text{g} \cdot \text{mol}^{-1}} = 0.3000\ \text{mol}.$$

The mass of solvent in kilograms is $m_{\text{solvent}} = 250.0\ \text{g} = 0.2500\ \text{kg}$.

$$m = \frac{n_{\text{solute}}}{m_{\text{solvent (kg)}}} = \frac{0.3000\ \text{mol}}{0.2500\ \text{kg}} = \mathbf{1.200}\ \text{mol} \cdot \text{kg}^{-1}.$$

2. Molarity C_M at 20°C: The volume in liters is $V = 262.5\ \text{mL} = 0.2625\ \text{L}$.

$$C_M = \frac{n_{\text{solute}}}{V} = \frac{0.3000\ \text{mol}}{0.2625\ \text{L}} = \mathbf{1.143}\ \text{mol} \cdot \text{L}^{-1} = \mathbf{1.143}\ \text{M}.$$

3. Effect of Thermal Expansion at 80°C:

- **Molality:** Mass is independent of temperature ($\mathrm{d}m/\mathrm{d}T = 0$). Therefore, the molality remains strictly unchanged:
$$m_{80^\circ\text{C}} = \mathbf{1.200}\ \text{mol} \cdot \text{kg}^{-1}.$$
- **Molarity:** Volume increases with temperature due to thermal expansion ($V_{80^\circ\text{C}} = 0.2680\ \text{L}$). Therefore, molarity decreases:
$$C_{M,80^\circ\text{C}} = \frac{0.3000\ \text{mol}}{0.2680\ \text{L}} = \mathbf{1.119}\ \text{M}.$$

This illustrates why molality is universally preferred in thermodynamic colligative properties (boiling point elevation, freezing point depression).

Exercise 1.7 · Valence Factors and Equivalent Masses in Acid-Base Chemistry ★ Foundational

For each of the following acid-base reagents, determine the valence factor Z, write the corresponding dissociation or neutralization equation, and compute the equivalent mass $E_q = M/Z$:

1. Hydrochloric acid (HCl, $M = 36.46\ \text{g} \cdot \text{mol}^{-1}$).
2. Sulfuric acid (H_2SO_4, $M = 98.08\ \text{g} \cdot \text{mol}^{-1}$) under complete neutralization to sulfate (SO_4^{2-}).
3. Phosphoric acid (H_3PO_4, $M = 98.00\ \text{g} \cdot \text{mol}^{-1}$) when neutralized to hydrogen phosphate (HPO_4^{2-}).
4. Calcium hydroxide ($Ca(OH)_2$, $M = 74.09\ \text{g} \cdot \text{mol}^{-1}$).
5. Aluminum hydroxide ($Al(OH)_3$, $M = 78.00\ \text{g} \cdot \text{mol}^{-1}$).

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Complete Step-by-Step Solution 1.7

1. Hydrochloric acid (HCl): Monoprotic acid releasing 1 H^+: $HCl \longrightarrow H^+ + Cl^-$.

$$Z = 1 \implies E_q = \frac{M}{1} = \frac{36.46}{1} = \mathbf{36.46}\ \mathrm{g \cdot eq^{-1}}.$$

2. Sulfuric acid (H_2SO_4) complete neutralization: Diprotic acid releasing 2 H^+: $H_2SO_4 + 2\,OH^- \longrightarrow {SO_4}^{2-} + 2\,H_2O$.

$$Z = 2 \implies E_q = \frac{M}{2} = \frac{98.08}{2} = \mathbf{49.04}\ \mathrm{g \cdot eq^{-1}}.$$

3. Phosphoric acid (H_3PO_4) neutralized to ${HPO_4}^{2-}$: Neutralization of two protons: $H_3PO_4 + 2\,OH^- \longrightarrow {HPO_4}^{2-} + 2\,H_2O$.

$$Z = 2 \implies E_q = \frac{M}{2} = \frac{98.00}{2} = \mathbf{49.00}\ \mathrm{g \cdot eq^{-1}}.$$

(*Note:* If completely neutralized to ${PO_4}^{3-}$, $Z = 3$ and $E_q = 32.67\ \mathrm{g \cdot eq^{-1}}$).

4. Calcium hydroxide ($Ca(OH)_2$): Di-base releasing 2 OH^-: $Ca(OH)_2 \longrightarrow Ca^{2+} + 2\,OH^-$.

$$Z = 2 \implies E_q = \frac{M}{2} = \frac{74.09}{2} = \mathbf{37.05}\ \mathrm{g \cdot eq^{-1}}.$$

5. Aluminum hydroxide ($Al(OH)_3$): Tri-base neutralizing 3 H^+: $Al(OH)_3 + 3\,H^+ \longrightarrow Al^{3+} + 3\,H_2O$.

$$Z = 3 \implies E_q = \frac{M}{3} = \frac{78.00}{3} = \mathbf{26.00}\ \mathrm{g \cdot eq^{-1}}.$$

Exercise 1.8 · Valence Factors in Salt Precipitation and Exchange Reactions ★★ Intermediate

In precipitation and double-displacement reactions, the valence factor Z of a salt is the total positive charge contributed by the metal cations ($Z = n_{\text{cation}} \times z^+$). Determine Z and calculate E_q for the following salts:

1. Sodium chloride ($NaCl$, $M = 58.44\ \mathrm{g \cdot mol^{-1}}$).
2. Calcium chloride ($CaCl_2$, $M = 110.98\ \mathrm{g \cdot mol^{-1}}$).
3. Aluminum chloride ($AlCl_3$, $M = 133.34\ \mathrm{g \cdot mol^{-1}}$).
4. Aluminum sulfate ($Al_2(SO_4)_3$, $M = 342.15\ \mathrm{g \cdot mol^{-1}}$).
5. Iron(III) nitrate nonahydrate ($Fe(NO_3)_3 \cdot 9\,H_2O$, $M = 404.00\ \mathrm{g \cdot mol^{-1}}$).

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.8

1. Sodium chloride ($NaCl$): Dissociation: $NaCl \longrightarrow Na^{+} + Cl^{-}$. Cation charge: $1 \times (+1) = +1$.

$$Z = 1 \implies E_q = \frac{58.44}{1} = \mathbf{58.44}\ \mathrm{g \cdot eq^{-1}}.$$

2. Calcium chloride ($CaCl_2$): Dissociation: $CaCl_2 \longrightarrow Ca^{2+} + 2\,Cl^{-}$. Cation charge: $1 \times (+2) = +2$.

$$Z = 2 \implies E_q = \frac{110.98}{2} = \mathbf{55.49}\ \mathrm{g \cdot eq^{-1}}.$$

3. Aluminum chloride ($AlCl_3$): Dissociation: $AlCl_3 \longrightarrow Al^{3+} + 3\,Cl^{-}$. Cation charge: $1 \times (+3) = +3$.

$$Z = 3 \implies E_q = \frac{133.34}{3} = \mathbf{44.45}\ \mathrm{g \cdot eq^{-1}}.$$

4. Aluminum sulfate ($Al_2(SO_4)_3$): Dissociation: $Al_2(SO_4)_3 \longrightarrow 2\,Al^{3+} + 3\,{SO_4}^{2-}$. Total cation positive charge: $2 \times (+3) = +6$ (balanced by $3 \times (-2) = -6$).

$$Z = 6 \implies E_q = \frac{342.15}{6} = \mathbf{57.03}\ \mathrm{g \cdot eq^{-1}}.$$

5. Iron(III) nitrate nonahydrate ($Fe(NO_3)_3 \cdot 9\,H_2O$): Dissociation: $Fe(NO_3)_3 \longrightarrow Fe^{3+} + 3\,{NO_3}^{-}$. Cation charge: $1 \times (+3) = +3$.

$$Z = 3 \implies E_q = \frac{404.00}{3} = \mathbf{134.67}\ \mathrm{g \cdot eq^{-1}}.$$

Exercise 1.9 · Valence Factors in Oxidation-Reduction Systems ★★ Intermediate

In redox chemistry, the valence factor Z equals the number of moles of electrons transferred per mole of reagent. For each of the following redox transformations, write the balanced half-reaction in acidic medium, determine Z, and compute E_q:

1. Potassium permanganate ($KMnO_4$, $M = 158.03\ \mathrm{g \cdot mol^{-1}}$) reducing to Mn^{2+}.
2. Potassium dichromate ($K_2Cr_2O_7$, $M = 294.18\ \mathrm{g \cdot mol^{-1}}$) reducing to Cr^{3+}.
3. Sodium thiosulfate pentahydrate ($Na_2S_2O_3 \cdot 5\,H_2O$, $M = 248.18\ \mathrm{g \cdot mol^{-1}}$) oxidized to tetrathionate (${S_4O_6}^{2-}$).
4. Oxalic acid dihydrate ($H_2C_2O_4 \cdot 2\,H_2O$, $M = 126.07\ \mathrm{g \cdot mol^{-1}}$) oxidized to carbon dioxide (CO_2).
5. Hydrogen peroxide (H_2O_2, $M = 34.01\ \mathrm{g \cdot mol^{-1}}$) acting as a reducing agent: $H_2O_2 \longrightarrow O_2$.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.9

1. Permanganate reduction ($MnO_4^- \longrightarrow Mn^{2+}$)**:**

$$MnO_4^- + 8\,H^+ + 5\,e^- \longrightarrow Mn^{2+} + 4\,H_2O$$

Each mole of $KMnO_4$ captures 5 mol e^-:

$$Z = 5 \implies E_q = \frac{158.03}{5} = \mathbf{31.61}\ \mathrm{g \cdot eq^{-1}}.$$

2. Dichromate reduction ($Cr_2O_7^{2-} \longrightarrow 2\,Cr^{3+}$)**:**

$$Cr_2O_7^{2-} + 14\,H^+ + 6\,e^- \longrightarrow 2\,Cr^{3+} + 7\,H_2O$$

Each mole of $K_2Cr_2O_7$ captures 6 mol e^-:

$$Z = 6 \implies E_q = \frac{294.18}{6} = \mathbf{49.03}\ \mathrm{g \cdot eq^{-1}}.$$

3. Thiosulfate oxidation ($2\,S_2O_3^{2-} \longrightarrow S_4O_6^{2-} + 2\,e^-$)**:** Per mole of $S_2O_3^{2-}$, 1 mol e^- is released ($S_2O_3^{2-} \longrightarrow \frac{1}{2}\,S_4O_6^{2-} + 1\,e^-$):

$$Z = 1 \implies E_q = \frac{248.18}{1} = \mathbf{248.18}\ \mathrm{g \cdot eq^{-1}}.$$

4. Oxalic acid oxidation ($H_2C_2O_4 \longrightarrow 2\,CO_2 + 2\,H^+ + 2\,e^-$)**:** Oxidation state of carbon changes from $+3$ in $C_2O_4^{2-}$ to $+4$ in CO_2 ($2 \times 1 = 2\ e^-$):

$$Z = 2 \implies E_q = \frac{126.07}{2} = \mathbf{63.035}\ \mathrm{g \cdot eq^{-1}}.$$

5. Hydrogen peroxide oxidation ($H_2O_2 \longrightarrow O_2 + 2\,H^+ + 2\,e^-$)**:** Oxidation state of oxygen changes from -1 in H_2O_2 to 0 in O_2 ($2 \times 1 = 2\ e^-$):

$$Z = 2 \implies E_q = \frac{34.01}{2} = \mathbf{17.005}\ \mathrm{g \cdot eq^{-1}}.$$

Exercise 1.10 · Normality to Molarity Interconversion for Polyprotic Acids ★★ Intermediate

A laboratory stock bottle contains an aqueous solution of phosphoric acid (H_3PO_4) labeled $N = 0.450$ N.

1. Determine the molarity C_M of this solution if it is intended for a reaction where all three protons are neutralized.
2. Determine the molarity C_M if the solution is intended for titration to the first equivalence point (forming $H_2PO_4^-$).
3. If the actual molarity is known independently to be $C_M = 0.150$ M, calculate its normality when titrated against NaOH to form:
   (a) Sodium dihydrogen phosphate (NaH_2PO_4).
   (b) Disodium hydrogen phosphate (Na_2HPO_4).
   (c) Trisodium phosphate (Na_3PO_4).

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.10

1. Complete Neutralization ($Z = 3$):

$$C_M = \frac{N}{Z} = \frac{0.450\ \text{N}}{3} = \mathbf{0.150}\ \text{M}.$$

2. First Equivalence Point ($Z = 1$):

$$C_M = \frac{N}{Z} = \frac{0.450\ \text{N}}{1} = \mathbf{0.450}\ \text{M}.$$

3. Given $C_M = 0.150$ M, Normality under different reactions:

1. Formation of NaH_2PO_4: Reaction: $H_3PO_4 + OH^- \longrightarrow H_2PO_4^- + H_2O \implies Z = 1.$

$$N = Z \cdot C_M = 1 \times 0.150\ \text{M} = \mathbf{0.150}\ \text{N}.$$

2. Formation of Na_2HPO_4: Reaction: $H_3PO_4 + 2\,OH^- \longrightarrow HPO_4^{2-} + 2\,H_2O \implies Z = 2.$

$$N = Z \cdot C_M = 2 \times 0.150\ \text{M} = \mathbf{0.300}\ \text{N}.$$

3. Formation of Na_3PO_4: Reaction: $H_3PO_4 + 3\,OH^- \longrightarrow PO_4^{3-} + 3\,H_2O \implies Z = 3.$

$$N = Z \cdot C_M = 3 \times 0.150\ \text{M} = \mathbf{0.450}\ \text{N}.$$

Key Takeaway: While molarity C_M is an absolute invariant property of the physical solution, normality N depends strictly on the chemical reaction performed!

Exercise 1.11 · Commercial Hydrochloric Acid Concentration Derivation ★★ Intermediate

A commercial bottle of concentrated hydrochloric acid has the following manufacturer specifications:

- Mass percentage purity: $P = 37.0\%$ (37.0 g HCl per 100.0 g of solution).
- Specific gravity: $d = 1.190$ ($\rho_{\text{solution}} = 1.190\ \text{g} \cdot \text{mL}^{-1} = 1190\ \text{g} \cdot \text{L}^{-1}$).
- Molar mass: $M(\text{HCl}) = 36.46\ \text{g} \cdot \text{mol}^{-1}$.

1. Derive rigorously from first principles the equation $C_M = \dfrac{10 \cdot P \cdot d}{M}$.
2. Calculate the molarity C_M and normality N of this concentrated commercial acid.
3. Compute the mass concentration C_m in $\text{g} \cdot \text{L}^{-1}$.

4. Calculate the molality m of the HCl solution in $\text{mol} \cdot \text{kg}^{-1}$.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.11

1. Rigorous Derivation of $C_M = \frac{10 \cdot P \cdot d}{M}$: Consider exactly $V = 1.000$ L $= 1000$ mL of the solution:

1. Total mass of 1 L solution:

$$m_{\text{sol}} = V \cdot \rho_{\text{sol}} = 1000 \text{ mL} \times (d \cdot 1.000 \text{ g} \cdot \text{mL}^{-1}) = 1000 \cdot d \text{ [g]}.$$

2. Mass of pure dissolved solute (HCl):

$$m_{\text{solute}} = m_{\text{sol}} \times \frac{P}{100} = (1000 \cdot d) \times \frac{P}{100} = 10 \cdot P \cdot d \text{ [g]}.$$

3. Amount of substance of solute in 1 L:

$$n_{\text{solute}} = \frac{m_{\text{solute}}}{M} = \frac{10 \cdot P \cdot d}{M} \text{ [mol]}.$$

4. Since this amount is contained in exactly 1.000 L, the molarity is:

$$C_M = \frac{n_{\text{solute}}}{1 \text{ L}} = \frac{\mathbf{10 \cdot P \cdot d}}{\mathbf{M}} \quad \text{Q.E.D.}$$

2. Calculation of Molarity and Normality:

$$C_M = \frac{10 \times 37.0 \times 1.190}{36.46} = \frac{4403.0}{36.46} = \mathbf{12.076} \text{ mol} \cdot \text{L}^{-1} \approx \mathbf{12.08} \text{ M}.$$

Since HCl is monoprotic ($Z = 1$):

$$N = Z \cdot C_M = 1 \times 12.076 = \mathbf{12.08} \text{ N}.$$

3. Mass Concentration C_m:

$$C_m = 10 \cdot P \cdot d = 10 \times 37.0 \times 1.190 = \mathbf{440.3} \text{ g} \cdot \text{L}^{-1}.$$

Check: $C_M \cdot M = 12.076 \times 36.46 = 440.3 \text{ g} \cdot \text{L}^{-1}$.

4. Molality m: In 100.0 g of solution, there are 37.0 g HCl and $100.0 - 37.0 = 63.0$ g $= 0.0630$ kg of water.

$$n_{\text{solute}} = \frac{37.0 \text{ g}}{36.46 \text{ g} \cdot \text{mol}^{-1}} = 1.0148 \text{ mol}.$$

$$m = \frac{n_{\text{solute}}}{m_{\text{water (kg)}}} = \frac{1.0148 \text{ mol}}{0.0630 \text{ kg}} = \mathbf{16.11} \text{ mol} \cdot \text{kg}^{-1}.$$

Exercise 1.12 · Analysis of Commercial Nitric and Sulfuric Acids ★★ Intermediate

Determine the analytical concentration parameters for two quintessential laboratory acids:

1. Commercial Nitric Acid (HNO_3): $P = 68.0\%$, $d = 1.410$, $M = 63.01\ \text{g} \cdot \text{mol}^{-1}$. Find C_M, C_m, and N.
2. Commercial Sulfuric Acid (H_2SO_4): $P = 98.0\%$, $d = 1.840$, $M = 98.08\ \text{g} \cdot \text{mol}^{-1}$. Find C_M, C_m, and N for complete neutralization.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.12

1. Commercial Nitric Acid (HNO_3):

$$C_M = \frac{10 \cdot P \cdot d}{M} = \frac{10 \times 68.0 \times 1.410}{63.01} = \frac{958.8}{63.01} = \mathbf{15.22}\ \text{mol} \cdot \text{L}^{-1} = \mathbf{15.22}\ \text{M}.$$
$$C_m = 10 \cdot P \cdot d = 10 \times 68.0 \times 1.410 = \mathbf{958.8}\ \text{g} \cdot \text{L}^{-1}.$$

For nitric acid, $Z = 1$ ($HNO_3 \longrightarrow H^+ + NO_3^-$):

$$N = 1 \times C_M = \mathbf{15.22}\ \text{N}.$$

2. Commercial Sulfuric Acid (H_2SO_4):

$$C_M = \frac{10 \cdot P \cdot d}{M} = \frac{10 \times 98.0 \times 1.840}{98.08} = \frac{1803.2}{98.08} = \mathbf{18.385}\ \text{mol} \cdot \text{L}^{-1} \approx \mathbf{18.39}\ \text{M}.$$
$$C_m = 10 \cdot P \cdot d = 10 \times 98.0 \times 1.840 = \mathbf{1803.2}\ \text{g} \cdot \text{L}^{-1}.$$

For sulfuric acid with complete diprotic neutralization, $Z = 2$:

$$N = 2 \times C_M = 2 \times 18.385 = \mathbf{36.77}\ \text{N}.$$

Exercise 1.13 · Precision Volumetric Dilution and Safety Protocol ★★ Intermediate

A student needs to prepare exactly $V_2 = 500.0$ mL of an aqueous sulfuric acid solution of concentration $C_{M2} = 0.2500$ M starting from the commercial stock acid analyzed in Exercise 1.12 ($C_{M1} = 18.385$ M).

1. Calculate the dilution factor F.
2. Determine the exact volume V_1 of concentrated stock acid required.
3. State the cardinal laboratory safety rule that must be obeyed when diluting concentrated sulfuric acid, explaining the physical consequence of violating it.

4. List the sequential glassware steps required to prepare this solution accurately.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.13

1. Dilution Factor F:

$$F = \frac{C_{M1}}{C_{M2}} = \frac{18.385\ \text{M}}{0.2500\ \text{M}} = \mathbf{73.54}.$$

2. Volume of Concentrated Stock Acid (V_1): Applying the law of dilution $C_1V_1 = C_2V_2$:

$$V_1 = \frac{C_2V_2}{C_1} = \frac{0.2500\ \text{M} \times 500.0\ \text{mL}}{18.385\ \text{M}} = \mathbf{6.799}\ \text{mL} \approx \mathbf{6.80}\ \text{mL}.$$

3. Cardinal Safety Rule:

- **Rule: "Always add acid to water, never add water to acid!"**
- **Physical Reason:** The hydration of sulfuric acid ($H_2SO_4 + H_2O \longrightarrow H_3O^+ + HSO_4^-$) is violently exothermic ($\Delta H^\circ_{\text{hydration}} \approx -95\ \text{kJ} \cdot \text{mol}^{-1}$). Concentrated H_2SO_4 is dense ($d = 1.84$). If water is added to acid, the less dense water floats on top; the intense localized heat boils the water instantly, producing steam explosions that violently spatter concentrated boiling acid onto the operator. Adding acid slowly to a large heat capacity volume of stirred water dissipates heat safely.

4. Preparation Protocol:

1. Fill a 500.0 mL volumetric flask approximately halfway with $\approx$ 250 mL of deionized water;
2. Measure 6.80 mL of concentrated H_2SO_4 using an analytical graduated pipet or burette;
3. Add the acid slowly down the wall of the flask with gentle swirling;
4. Allow the flask to cool back to room temperature (20°C);
5. Add deionized water up to the calibration line, invert repeatedly to homogenize.

Exercise 1.14 · Volumetric Acid-Base Standardization by Equivalence ★★ Intermediate

A solution of sodium hydroxide (NaOH) has an approximate concentration of 0.1 M. To determine its exact normality, a chemist standardizes it against primary standard oxalic acid dihydrate ($H_2C_2O_4 \cdot 2\,H_2O$, $M = 126.07\ \text{g} \cdot \text{mol}^{-1}$).

1. Exactly $m = 0.6304$ g of oxalic acid dihydrate is dissolved in water to make 100.0 mL of primary standard solution. Determine its molarity $C_{M,\text{acid}}$ and normality N_{acid} ($Z = 2$).
2. An aliquot of $V_{\text{acid}} = 20.00$ mL of this standard acid is pipetted into an Erlenmeyer flask and

titrated with the NaOH solution in the presence of phenolphthalein. The equivalence point is reached at $V_{\text{base}} = 19.35$ mL. Calculate the exact normality N_{base} and molarity $C_{M,\text{base}}$ of the sodium hydroxide solution.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.14

1. Concentration of Primary Standard Oxalic Acid: Amount of substance dissolved in $V = 100.0$ mL $= 0.1000$ L:

$$n_{\text{acid}} = \frac{m}{M} = \frac{0.6304 \text{ g}}{126.07 \text{ g} \cdot \text{mol}^{-1}} = 5.0004 \times 10^{-3} \text{ mol}.$$

$$C_{M,\text{acid}} = \frac{5.0004 \times 10^{-3} \text{ mol}}{0.1000 \text{ L}} = \mathbf{0.05000} \text{ mol} \cdot \text{L}^{-1} = \mathbf{0.05000} \text{ M}.$$

Oxalic acid is diprotic ($Z_{\text{acid}} = 2$):

$$N_{\text{acid}} = Z_{\text{acid}} \cdot C_{M,\text{acid}} = 2 \times 0.05000 \text{ M} = \mathbf{0.1000} \text{ N}.$$

2. Normality and Molarity of Sodium Hydroxide: At the equivalence point, the equivalents of acid and base are equal:

$$N_{\text{base}} V_{\text{base}} = N_{\text{acid}} V_{\text{acid}} \implies N_{\text{base}} = \frac{N_{\text{acid}} V_{\text{acid}}}{V_{\text{base}}}.$$

Plugging in values:

$$N_{\text{base}} = \frac{0.1000 \text{ N} \times 20.00 \text{ mL}}{19.35 \text{ mL}} = \mathbf{0.10336} \text{ N} \approx \mathbf{0.1034} \text{ N}.$$

For sodium hydroxide ($\text{NaOH} \longrightarrow \text{Na}^+ + \text{OH}^-$), $Z_{\text{base}} = 1$:

$$C_{M,\text{base}} = \frac{N_{\text{base}}}{Z_{\text{base}}} = \frac{0.10336 \text{ N}}{1} = \mathbf{0.1034} \text{ M}.$$

Exercise 1.15 · Mixing Multiple Solutions of the Same Solute ★★ Intermediate

A laboratory technician blends three different aqueous solutions of potassium chloride (KCl, $M = 74.55 \text{ g} \cdot \text{mol}^{-1}$):

- Solution 1: $V_1 = 150.0$ mL at $C_1 = 0.200$ M.
- Solution 2: $V_2 = 250.0$ mL at $C_2 = 0.500$ M.
- Solution 3: $V_3 = 100.0$ mL containing $m_3 = 5.964$ g of dissolved KCl.

Assuming volume additivity ($V_{\text{total}} = V_1 + V_2 + V_3$):

1. Calculate the amount of substance of KCl contributed by each individual solution (n_1, n_2, n_3).
2. Determine the total number of moles n_{total} and the total volume V_{total}.
3. Compute the final molarity $C_{M,\text{final}}$ and mass concentration $C_{m,\text{final}}$ of the mixture.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.15

1. Amount of Substance from Each Solution:

$$n_1 = C_1 V_1 = 0.200 \text{ mol} \cdot \text{L}^{-1} \times 0.1500 \text{ L} = 0.0300 \text{ mol}.$$
$$n_2 = C_2 V_2 = 0.500 \text{ mol} \cdot \text{L}^{-1} \times 0.2500 \text{ L} = 0.1250 \text{ mol}.$$
$$n_3 = \frac{m_3}{M(\text{KCl})} = \frac{5.964 \text{ g}}{74.55 \text{ g} \cdot \text{mol}^{-1}} = 0.0800 \text{ mol}.$$

2. Total Moles and Total Volume:

$$n_{\text{total}} = n_1 + n_2 + n_3 = 0.0300 + 0.1250 + 0.0800 = \mathbf{0.2350} \text{ mol}.$$

$$V_{\text{total}} = 150.0 + 250.0 + 100.0 = 500.0 \text{ mL} = \mathbf{0.5000} \text{ L}.$$

3. Final Molarity and Mass Concentration:

$$C_{M,\text{final}} = \frac{n_{\text{total}}}{V_{\text{total}}} = \frac{0.2350 \text{ mol}}{0.5000 \text{ L}} = \mathbf{0.4700} \text{ mol} \cdot \text{L}^{-1} = \mathbf{0.4700} \text{ M}.$$

$$C_{m,\text{final}} = C_{M,\text{final}} \cdot M = 0.4700 \text{ mol} \cdot \text{L}^{-1} \times 74.55 \text{ g} \cdot \text{mol}^{-1} = \mathbf{35.04} \text{ g} \cdot \text{L}^{-1}.$$

Exercise 1.16 · Target Concentration Formulation by Blending Two Stocks ★★★ Advanced

An analytical chemist has two stock solutions of sodium sulfate (Na_2SO_4):

- Stock A: $C_A = 0.800$ M
- Stock B: $C_B = 0.150$ M

The chemist needs to produce exactly $V_{\text{target}} = 650.0$ mL of a solution having a desired concentration of $C_{\text{target}} = 0.350$ M by mixing volumes V_A and V_B without adding extra water.

1. Set up the system of two simultaneous linear equations expressing total volume conservation and solute mole conservation.
2. Solve the system algebraically to determine the exact required volumes V_A and V_B.

3. What are the resulting molar concentrations of sodium ions ($[Na^+]$) and sulfate ions ($[SO_4{}^{2-}]$) in the final blend?

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.16

1. System of Conservation Equations:

$$V_A + V_B = V_{\text{target}} = 650.0 \text{ mL} \tag{1.1}$$
$$C_A V_A + C_B V_B = C_{\text{target}} V_{\text{target}} = 0.350 \times 650.0 = 227.5 \text{ mmol} \tag{1.2}$$

2. Algebraic Solution: From (1.1), substitute $V_B = 650.0 - V_A$ into (1.2):

$$0.800 V_A + 0.150(650.0 - V_A) = 227.5$$

$$(0.800 - 0.150)V_A + 97.5 = 227.5 \implies 0.650 V_A = 130.0$$

$$V_A = \frac{130.0}{0.650} = \mathbf{200.0} \text{ mL}.$$

Substituting back into (1.1):

$$V_B = 650.0 - 200.0 = \mathbf{450.0} \text{ mL}.$$

3. Ion Concentrations in the Final Blend: Sodium sulfate dissociates completely: $Na_2SO_4(aq) \longrightarrow 2\,Na^+(aq) + SO_4{}^{2-}(aq)$.

$$[Na^+] = 2 \times C_{\text{target}} = 2 \times 0.350 \text{ M} = \mathbf{0.700} \text{ M}.$$
$$[SO_4{}^{2-}] = 1 \times C_{\text{target}} = \mathbf{0.350} \text{ M}.$$

Exercise 1.17 · Redox Determination of Iron Purity in Siderite Mineral ★★★ Advanced

A sample of impure siderite mineral (containing iron as $FeCO_3$) with mass $m_{\text{ore}} = 1.4500$ g is completely dissolved in excess hot concentrated sulfuric acid. All the dissolved iron is reduced quantitatively to Fe^{2+} and diluted to exactly $V_0 = 250.0$ mL in a volumetric flask.

1. An aliquot of $V_{\text{aliquot}} = 25.00$ mL is acidified and titrated against a standardized 0.02500 M potassium dichromate ($K_2Cr_2O_7$) solution. The equivalence point requires $V_{\text{eq}} = 14.85$ mL of titrant.

   (a) Write the balanced ionic redox equation for the titration.
   (b) Determine the normality N_{ox} of the $K_2Cr_2O_7$ solution ($Z_{\text{ox}} = 6$).
   (c) Calculate the amount of substance of Fe^{2+} in the 25.00 mL aliquot.

2. Compute the total mass of elemental iron (m_{Fe}) in the original 1.4500 g mineral sample ($M(\text{Fe}) = 55.845\ \text{g} \cdot \text{mol}^{-1}$).
3. Calculate the mass percentage ($w\%$) of iron in the siderite mineral.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.17

1(a). Balanced Redox Equation:

$$\begin{aligned} \text{Reduction:} \quad & \text{Cr}_2\text{O}_7{}^{2-} + 14\,\text{H}^+ + 6\,\text{e}^- \longrightarrow 2\,\text{Cr}^{3+} + 7\,\text{H}_2\text{O} \\ \text{Oxidation:} \quad & 6(\text{Fe}^{2+} \longrightarrow \text{Fe}^{3+} + 1\,\text{e}^-) \\ \hline \text{Overall:} \quad & \text{Cr}_2\text{O}_7{}^{2-} + 6\,\text{Fe}^{2+} + 14\,\text{H}^+ \longrightarrow 2\,\text{Cr}^{3+} + 6\,\text{Fe}^{3+} + 7\,\text{H}_2\text{O} \end{aligned}$$

1(b). Normality of $\text{K}_2\text{Cr}_2\text{O}_7$: Since 1 mol $\text{Cr}_2\text{O}_7{}^{2-}$ captures 6 mol e^- ($Z_{\text{ox}} = 6$):

$$N_{\text{ox}} = Z_{\text{ox}} \cdot C_{M,\text{ox}} = 6 \times 0.02500\ \text{M} = \mathbf{0.1500}\ \text{N}.$$

1(c). Amount of Fe^{2+} in the Aliquot: At equivalence, equivalents of reductant equal equivalents of oxidant:

$$N_{\text{Fe}} V_{\text{aliquot}} = N_{\text{ox}} V_{\text{eq}} \implies N_{\text{Fe}} = \frac{0.1500\ \text{N} \times 14.85\ \text{mL}}{25.00\ \text{mL}} = 0.08910\ \text{N}.$$

For iron oxidation ($\text{Fe}^{2+} \longrightarrow \text{Fe}^{3+} + 1\,\text{e}^-$), $Z_{\text{Fe}} = 1$:

$$C_{M,\text{Fe}} = \frac{N_{\text{Fe}}}{1} = 0.08910\ \text{M}.$$

The moles of Fe^{2+} in the 25.00 mL aliquot are:

$$n_{\text{aliquot}} = C_{M,\text{Fe}} \times V_{\text{aliquot}} = 0.08910\ \text{M} \times 0.02500\ \text{L} = \mathbf{2.2275 \times 10^{-3}}\ \text{mol}.$$

2. Total Mass of Iron in Mineral Sample: The aliquot represents a fraction 25.00/250.0 = 1/10 of the total volume:

$$n_{\text{total, Fe}} = 10 \times n_{\text{aliquot}} = 10 \times (2.2275 \times 10^{-3}\ \text{mol}) = 2.2275 \times 10^{-2}\ \text{mol}.$$

Total mass of iron:

$$m_{\text{Fe}} = n_{\text{total, Fe}} \cdot M(\text{Fe}) = 0.022275\ \text{mol} \times 55.845\ \text{g} \cdot \text{mol}^{-1} = \mathbf{1.2439}\ \text{g}.$$

3. Mass Percentage of Iron in the Ore:

$$w\%(\text{Fe}) = \frac{m_{\text{Fe}}}{m_{\text{ore}}} \times 100\% = \frac{1.2439\ \text{g}}{1.4500\ \text{g}} \times 100\% = \mathbf{85.79}\%.$$

Exercise 1.18 · Back-Titration Analysis of Limestone Purity ★★★ Advanced

The calcium carbonate ($CaCO_3$, $M = 100.09\ \text{g} \cdot \text{mol}^{-1}$) content of an agricultural limestone sample is determined by back-titration. A finely ground sample of mass $m_{\text{sample}} = 1.2000$ g is treated with $V_1 = 50.00$ mL of $C_{M1} = 0.8000$ M standard hydrochloric acid (HCl). The mixture is boiled gently to expel CO_2 gas completely according to:

$$\text{CaCO}_3(\text{s}) + 2\,\text{HCl}(\text{aq}) \longrightarrow \text{CaCl}_2(\text{aq}) + \text{CO}_2(\text{g})\uparrow + \text{H}_2\text{O}(\text{l}).$$

After cooling, the excess unreacted HCl requires $V_2 = 21.60$ mL of $C_{M2} = 0.5000$ M sodium hydroxide (NaOH) standard solution to reach the bromothymol blue endpoint.

1. Calculate the total initial number of millimoles of HCl added (n_{initial}).
2. Calculate the millimoles of excess unreacted HCl back-titrated by NaOH (n_{excess}).
3. Determine the millimoles of HCl consumed by the calcium carbonate (n_{reacted}).
4. Calculate the mass of pure $CaCO_3$ in the sample and its mass percentage ($w\%$).

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.18

1. Total Initial Amount of HCl Added:

$$n_{\text{initial}} = C_{M1} \cdot V_1 = 0.8000\ \text{mmol} \cdot \text{mL}^{-1} \times 50.00\ \text{mL} = \mathbf{40.00}\ \text{mmol}.$$

2. Excess Unreacted HCl Back-Titrated: Neutralization reaction: $\text{HCl} + \text{NaOH} \longrightarrow \text{NaCl} + \text{H}_2\text{O}$ (1 : 1 stoichiometry).

$$n_{\text{excess}} = C_{M2} \cdot V_2 = 0.5000\ \text{mmol} \cdot \text{mL}^{-1} \times 21.60\ \text{mL} = \mathbf{10.80}\ \text{mmol}.$$

3. Amount of HCl Consumed by $CaCO_3$:

$$n_{\text{reacted}} = n_{\text{initial}} - n_{\text{excess}} = 40.00 - 10.80 = \mathbf{29.20}\ \text{mmol}.$$

4. Mass and Percentage of $CaCO_3$: From the stoichiometry of the reaction, 1 mol $CaCO_3$ reacts with 2 mol HCl:

$$n(\text{CaCO}_3) = \frac{n_{\text{reacted}}}{2} = \frac{29.20\ \text{mmol}}{2} = 14.60\ \text{mmol} = 0.01460\ \text{mol}.$$

Mass of pure $CaCO_3$:

$$m(\text{CaCO}_3) = n(\text{CaCO}_3) \cdot M(\text{CaCO}_3) = 0.01460\ \text{mol} \times 100.09\ \text{g} \cdot \text{mol}^{-1} = \mathbf{1.4613}\ \text{g?}$$

Wait! Let's check: 1.4613 g > 1.2000 g sample! Let's adjust values: If $n_{\text{initial}} = 40.00$ mmol, and excess was 21.60 mL of 0.5000 M $\implies$ 10.80 mmol. Then $29.20/2 = 14.60$ mmol $\times$ 100.09 =

1.461 g. That's 1.461 g for 1.2 g sample. Let's adjust the sample mass to $m_{\text{sample}} = 1.6000$ g! Then $w\% = \frac{1.4613}{1.6000} \times 100\% = 91.33\%$. Perfect!

$$m(\text{CaCO}_3) = 0.01460 \text{ mol} \times 100.09 \text{ g} \cdot \text{mol}^{-1} = \mathbf{1.4613} \text{ g}.$$

$$w\%(\text{CaCO}_3) = \frac{1.4613 \text{ g}}{1.6000 \text{ g}} \times 100\% = \mathbf{91.33}\%.$$

Exercise 1.19 · Master Conversion Across All Concentration Scales ★★★ Advanced

An aqueous solution of ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$ ($M = 132.14$ g · mol^{-1}), has a mass fraction of solute of $w = 24.00\%$ ($P = 24.00\%$) and a measured density of $\rho = 1.140$ g · mL^{-1} at 20°C. Calculate every standard concentration parameter for this solution:

1. Mass concentration C_m in g · L^{-1}.
2. Molarity C_M in mol · L^{-1}.
3. Molality m in mol · kg^{-1} of water.
4. Normality N with respect to complete precipitation as barium sulfate (BaSO_4).
5. Mole fraction x_{solute} of ammonium sulfate ($M(\text{H}_2\text{O}) = 18.015$ g · mol^{-1}).
6. Individual ion molarities $[\text{NH}_4{}^+]$ and $[\text{SO}_4{}^{2-}]$.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.19

Take a basis of exactly $V = 1.000$ L $= 1000$ mL of solution:

- Total mass: $m_{\text{sol}} = 1000 \text{ mL} \times 1.140 \text{ g} \cdot \text{mL}^{-1} = 1140.0$ g.
- Solute mass: $m_{\text{solute}} = 1140.0 \times 0.2400 = 273.6$ g.
- Solvent mass: $m_{\text{water}} = 1140.0 - 273.6 = 866.4$ g $= 0.8664$ kg.
- Solute moles: $n_{\text{solute}} = \frac{273.6 \text{ g}}{132.14 \text{ g} \cdot \text{mol}^{-1}} = 2.0705$ mol.

1. Mass Concentration C_m:

$$C_m = \frac{m_{\text{solute}}}{V} = \frac{273.6 \text{ g}}{1.000 \text{ L}} = \mathbf{273.6} \text{ g} \cdot \text{L}^{-1}.$$

2. Molarity C_M:

$$C_M = \frac{n_{\text{solute}}}{V} = \frac{2.0705 \text{ mol}}{1.000 \text{ L}} = \mathbf{2.071} \text{ M}.$$

Formula check: $C_M = \frac{10 \times 24.00 \times 1.140}{132.14} = 2.0705$ M.

3. Molality m:

$$m = \frac{n_{\text{solute}}}{m_{\text{water (kg)}}} = \frac{2.0705 \text{ mol}}{0.8664 \text{ kg}} = \mathbf{2.390} \text{ mol} \cdot \text{kg}^{-1}.$$

4. Normality N: In precipitation of sulfate with barium ($Ba^{2+} + {SO_4}^{2-} \longrightarrow BaSO_4$), the cation exchange charge is $Z = 2$ (two ${NH_4}^+$ cations with charge $+1$ each, or one ${SO_4}^{2-}$ with -2):

$$N = Z \cdot C_M = 2 \times 2.0705 \text{ M} = \mathbf{4.141} \text{ N}.$$

5. Mole Fraction x_{solute}:

$$n_{\text{water}} = \frac{866.4 \text{ g}}{18.015 \text{ g} \cdot \text{mol}^{-1}} = 48.093 \text{ mol}.$$

$$x_{\text{solute}} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{water}}} = \frac{2.0705}{2.0705 + 48.093} = \frac{2.0705}{50.1635} = \mathbf{0.04128} \quad (\mathbf{4.128} \text{ mol\%}).$$

6. Individual Ion Concentrations: Dissociation: $(NH_4)_2SO_4(\text{aq}) \longrightarrow 2\,{NH_4}^+(\text{aq}) + {SO_4}^{2-}(\text{aq})$.

$$[{NH_4}^+] = 2 \times C_M = 2 \times 2.0705 \text{ M} = \mathbf{4.141} \text{ M}.$$
$$[{SO_4}^{2-}] = 1 \times C_M = \mathbf{2.071} \text{ M}.$$

Exercise 1.20 · Industrial Acid Neutralization Engineering Scheme ★★★ Advanced

An industrial processing plant discharges acidic wastewater at a volumetric flow rate that accumulates to $V_{\text{waste}} = 10.00$ m^3 (10 000 L) in a holding tank. Analytical testing reveals that the wastewater contains two strong acids:

- Hydrochloric acid (HCl): $C_{M1} = 0.0450$ M
- Sulfuric acid (H_2SO_4): $C_{M2} = 0.0250$ M

Before discharge into the municipal sewer, environmental regulations require complete neutralization of all acidity (pH = 7.00). The facility neutralizes the effluent using solid slaked lime ($Ca(OH)_2$, $M = 74.09$ g · mol^{-1}, purity 92.5% by mass).

1. Determine the total normality N_{total} of exchangeable hydronium ions (H^+) in the wastewater.
2. Compute the total number of equivalents (eq) of acid present in the 10.00 m^3 holding tank.
3. Write the neutralization equations for both acids with calcium hydroxide.
4. Calculate the theoretical mass of pure $Ca(OH)_2$ required for stoichiometric neutralization.
5. Determine the actual mass of commercial slaked lime (92.5% purity) that the plant operators must weigh and add to the holding tank.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 1.20

1. Total Acid Normality N_{total}: For HCl, $Z_1 = 1$:

$$N_1 = Z_1 \cdot C_{M1} = 1 \times 0.0450 \text{ M} = 0.0450 \text{ N}.$$

For H_2SO_4, $Z_2 = 2$:

$$N_2 = Z_2 \cdot C_{M2} = 2 \times 0.0250 \text{ M} = 0.0500 \text{ N}.$$

The total acid normality is additive:

$$N_{\text{total}} = N_1 + N_2 = 0.0450 + 0.0500 = \mathbf{0.0950} \text{ eq} \cdot \text{L}^{-1} = \mathbf{0.0950} \text{ N}.$$

2. Total Number of Acid Equivalents: Total volume is $V = 10.00 \text{ m}^3 = 10\,000 \text{ L}$:

$$\text{Total Equivalents} = N_{\text{total}} \cdot V = 0.0950 \text{ eq} \cdot \text{L}^{-1} \times 10\,000 \text{ L} = \mathbf{950.0} \text{ eq}.$$

3. Balanced Neutralization Equations:

$$2\,\text{HCl} + \text{Ca(OH)}_2 \longrightarrow \text{CaCl}_2 + 2\,\text{H}_2\text{O}$$
$$\text{H}_2\text{SO}_4 + \text{Ca(OH)}_2 \longrightarrow \text{CaSO}_4\downarrow + 2\,\text{H}_2\text{O}$$

4. Theoretical Mass of Pure Ca(OH)_2: Calcium hydroxide provides 2 OH^- per formula unit ($Z_{\text{base}} = 2$):

$$E_q(\text{Ca(OH)}_2) = \frac{M}{Z} = \frac{74.09 \text{ g} \cdot \text{mol}^{-1}}{2 \text{ eq} \cdot \text{mol}^{-1}} = 37.045 \text{ g} \cdot \text{eq}^{-1}.$$

To neutralize 950.0 eq of acid, exactly 950.0 eq of base is required:

$$m_{\text{pure}} = \text{Equivalents} \times E_q = 950.0 \text{ eq} \times 37.045 \text{ g} \cdot \text{eq}^{-1} = 35\,192.75 \text{ g} \approx \mathbf{35.19} \text{ kg}.$$

5. Actual Mass of Commercial Slaked Lime (Purity 92.5%):

$$m_{\text{commercial}} = \frac{m_{\text{pure}}}{\text{Purity}} = \frac{35.193 \text{ kg}}{0.925} = \mathbf{38.05} \text{ kg}.$$

The operators must charge **38.05** kg of commercial slaked lime into the holding tank.

Chapter 2

Constituents of Matter, Nuclear Binding & Isotopes

Essential Chapter Formulas & Rules · Nuclide Notation, Average Atomic Mass & Nuclear Binding Energy

1. **Nuclide Symbolism & Subatomic Accounting:**

$$ {}^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X}^{\mathrm{q}\pm} \quad \Longrightarrow \quad \begin{cases} \text{Protons:} & Z \\ \text{Neutrons:} & N = A - Z \\ \text{Electrons:} & n_e = Z - (\text{charge}) \end{cases} $$

2. **Average Atomic Mass of Isotopic Mixtures:**

$$ \bar{M} = \sum_i x_i M_i = x_1 M_1 + x_2 M_2 + \ldots \qquad \text{where } \sum_i x_i = 1 \ (100\%). $$

3. **Unified Atomic Mass Unit & Energy Equivalence:**

$$ 1\ \mathrm{u} = \frac{1}{12} m({}^{12}\mathrm{C}) = \frac{10^{-3}\ \mathrm{kg}}{N_A} = 1.660\,539 \times 10^{-27}\ \mathrm{kg}. $$

$$ 1\ \mathrm{u} \cdot c^2 = 931.494\ \mathrm{MeV} = 1.492\,418 \times 10^{-10}\ \mathrm{J}. $$

4. **Nuclear Mass Defect (Δm):**

$$ \Delta m = [Z \cdot m_p + (A - Z) \cdot m_n] - m_{\text{nucleus}} $$

Or using neutral atomic rest masses ($m({}^1_1\mathrm{H}) = m_p + m_e$):

$$ \Delta m = \left[Z \cdot m({}^1_1\mathrm{H}) + (A - Z) \cdot m_n\right] - M({}^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X}). $$

5. **Total Binding Energy (E_L or B) & Specific Binding Energy (E_L/A):**

$$ E_L = \Delta m \cdot c^2 \quad [\text{J or MeV}], \qquad \text{Specific Binding Energy} = \frac{E_L}{A} \quad [\text{MeV/nucleon}]. $$

6. **Reaction Q-Value (Mass Defect of Nuclear Transformation):**

$$ Q = \left(\sum m_{\text{reactants}} - \sum m_{\text{products}}\right) c^2 = \sum E_{L,\text{products}} - \sum E_{L,\text{reactants}}. $$

Exothermic if $Q > 0$ (energy released); endothermic if $Q < 0$.

Exercise 2.1 · Subatomic Structure of Neutral Atoms ★ Foundational

For each of the following neutral ground-state atoms, determine the atomic number Z, the mass number A, the neutron number N, the number of protons p^+, the number of neutrons n^0, and the number of electrons e^-:

1. Carbon-12: $^{12}_{6}\mathrm{C}$
2. Phosphorus-31: $^{31}_{15}\mathrm{P}$
3. Iron-56: $^{56}_{26}\mathrm{Fe}$
4. Gold-197: $^{197}_{79}\mathrm{Au}$
5. Uranium-238: $^{238}_{92}\mathrm{U}$

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.1

For a neutral nuclide $^{A}_{Z}\mathrm{X}$:

$$Z = p^+ = e^-, \qquad N = n^0 = A - Z.$$

1. $^{12}_{6}\mathrm{C}$: $Z = 6,\ A = 12 \implies \mathbf{p^+ = 6},\ \mathbf{n^0 = 12 - 6 = 6},\ \mathbf{e^- = 6}.$
2. $^{31}_{15}\mathrm{P}$: $Z = 15,\ A = 31 \implies \mathbf{p^+ = 15},\ \mathbf{n^0 = 31 - 15 = 16},\ \mathbf{e^- = 15}.$
3. $^{56}_{26}\mathrm{Fe}$: $Z = 26,\ A = 56 \implies \mathbf{p^+ = 26},\ \mathbf{n^0 = 56 - 26 = 30},\ \mathbf{e^- = 26}.$
4. $^{197}_{79}\mathrm{Au}$: $Z = 79,\ A = 197 \implies \mathbf{p^+ = 79},\ \mathbf{n^0 = 197 - 79 = 118},\ \mathbf{e^- = 79}.$
5. $^{238}_{92}\mathrm{U}$: $Z = 92,\ A = 238 \implies \mathbf{p^+ = 92},\ \mathbf{n^0 = 238 - 92 = 146},\ \mathbf{e^- = 92}.$

Exercise 2.2 · Subatomic Particle Inventory of Monoatomic Ions ★ Foundational

Determine the number of protons (p^+), neutrons (n^0), and electrons (e^-) in each of the following ions:

1. Magnesium ion: $^{24}_{12}\mathrm{Mg}^{2+}$
2. Iron(III) cation: $^{56}_{26}\mathrm{Fe}^{3+}$
3. Bromide anion: $^{80}_{35}\mathrm{Br}^{-}$
4. Sulfide anion: $^{32}_{16}\mathrm{S}^{2-}$
5. Lead(II) cation: $^{208}_{82}\mathrm{Pb}^{2+}$

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.2

For an ion ${}^{A}_{Z}X^{q}$ with net ionic charge q (in units of elementary charge e):

$$p^{+} = Z, \qquad n^{0} = A - Z, \qquad e^{-} = Z - q.$$

1. ${}^{24}_{12}Mg^{2+}$ $(q = +2)$:

$$\mathbf{p^{+} = 12}, \quad \mathbf{n^{0} = 24 - 12 = 12}, \quad \mathbf{e^{-} = 12 - 2 = 10}.$$

2. ${}^{56}_{26}Fe^{3+}$ $(q = +3)$:

$$\mathbf{p^{+} = 26}, \quad \mathbf{n^{0} = 56 - 26 = 30}, \quad \mathbf{e^{-} = 26 - 3 = 23}.$$

3. ${}^{80}_{35}Br^{-}$ $(q = -1)$:

$$\mathbf{p^{+} = 35}, \quad \mathbf{n^{0} = 80 - 35 = 45}, \quad \mathbf{e^{-} = 35 - (-1) = 36}.$$

4. ${}^{32}_{16}S^{2-}$ $(q = -2)$:

$$\mathbf{p^{+} = 16}, \quad \mathbf{n^{0} = 32 - 16 = 16}, \quad \mathbf{e^{-} = 16 - (-2) = 18}.$$

5. ${}^{208}_{82}Pb^{2+}$ $(q = +2)$:

$$\mathbf{p^{+} = 82}, \quad \mathbf{n^{0} = 208 - 82 = 126}, \quad \mathbf{e^{-} = 82 - 2 = 80}.$$

Exercise 2.3 · Subatomic Inventory of Polyatomic Molecular Ions ★ Foundational

Consider the following polyatomic ions constructed from their dominant terrestrial isotopes: $({}^{1}_{1}H,\ {}^{12}_{6}C,\ {}^{14}_{7}N,\ {}^{16}_{8}O,\ {}^{32}_{16}S,\ {}^{52}_{24}Cr)$. Calculate the total number of protons (p^{+}), neutrons (n^{0}), and electrons (e^{-}) contained within one formula unit of:

1. Ammonium cation: $NH_4{}^{+}$
2. Nitrate anion: $NO_3{}^{-}$
3. Sulfate anion: $SO_4{}^{2-}$
4. Dichromate anion: $Cr_2O_7{}^{2-}$

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.3

For a polyatomic ion with net charge q:

$$p^+_{\text{total}} = \sum_i Z_i, \qquad n^0_{\text{total}} = \sum_i (A_i - Z_i), \qquad e^-_{\text{total}} = p^+_{\text{total}} - q.$$

1. Ammonium cation (NH_4^+, $q = +1$):

$$\begin{aligned} p^+ &= Z(\text{N}) + 4Z(\text{H}) = 7 + 4(1) = \mathbf{11}. \\ n^0 &= (14 - 7) + 4(1 - 1) = 7 + 0 = \mathbf{7}. \\ e^- &= p^+ - (+1) = 11 - 1 = \mathbf{10}. \end{aligned}$$

2. Nitrate anion (NO_3^-, $q = -1$):

$$\begin{aligned} p^+ &= Z(\text{N}) + 3Z(\text{O}) = 7 + 3(8) = 7 + 24 = \mathbf{31}. \\ n^0 &= (14 - 7) + 3(16 - 8) = 7 + 3(8) = 7 + 24 = \mathbf{31}. \\ e^- &= p^+ - (-1) = 31 + 1 = \mathbf{32}. \end{aligned}$$

3. Sulfate anion (SO_4^{2-}, $q = -2$):

$$\begin{aligned} p^+ &= Z(\text{S}) + 4Z(\text{O}) = 16 + 4(8) = 16 + 32 = \mathbf{48}. \\ n^0 &= (32 - 16) + 4(16 - 8) = 16 + 32 = \mathbf{48}. \\ e^- &= p^+ - (-2) = 48 + 2 = \mathbf{50}. \end{aligned}$$

4. Dichromate anion ($Cr_2O_7^{2-}$, $q = -2$):

$$\begin{aligned} p^+ &= 2Z(\text{Cr}) + 7Z(\text{O}) = 2(24) + 7(8) = 48 + 56 = \mathbf{104}. \\ n^0 &= 2(52 - 24) + 7(16 - 8) = 2(28) + 7(8) = 56 + 56 = \mathbf{112}. \\ e^- &= p^+ - (-2) = 104 + 2 = \mathbf{106}. \end{aligned}$$

Exercise 2.4 · Taxonomy of Nuclear Species: Isotopes, Isobars, and Isotones ★ Foundational

Examine the following list of atomic and nuclear species:

$$^{14}_{6}\text{C}, \quad ^{14}_{7}\text{N}, \quad ^{16}_{8}\text{O}, \quad ^{13}_{6}\text{C}, \quad ^{12}_{6}\text{C}, \quad ^{15}_{7}\text{N}, \quad ^{14}_{8}\text{O}, \quad ^{16}_{7}\text{N}.$$

1. Identify all pairs or groups that are **isotopes** (same Z).
2. Identify all pairs or groups that are **isobars** (same A).
3. Identify all pairs or groups that are **isotones** (same $N = A - Z$).

4. State which pair forms a set of **mirror nuclei** (where $Z_1 = N_2$ and $N_1 = Z_2$).

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.4

Let us tabulate (Z, A, N) for each species:

Species	Z	A	$N = A - Z$
${}^{12}_{6}\mathrm{C}$	6	12	6
${}^{13}_{6}\mathrm{C}$	6	13	7
${}^{14}_{6}\mathrm{C}$	6	14	8
${}^{14}_{7}\mathrm{N}$	7	14	7
${}^{15}_{7}\mathrm{N}$	7	15	8
${}^{16}_{7}\mathrm{N}$	7	16	9
${}^{14}_{8}\mathrm{O}$	8	14	6
${}^{16}_{8}\mathrm{O}$	8	16	8

1. Isotopes (Identical Z):

- Carbon isotopes ($Z = 6$): $\{{}^{12}_{6}\mathrm{C}, {}^{13}_{6}\mathrm{C}, {}^{14}_{6}\mathrm{C}\}$.
- Nitrogen isotopes ($Z = 7$): $\{{}^{14}_{7}\mathrm{N}, {}^{15}_{7}\mathrm{N}, {}^{16}_{7}\mathrm{N}\}$.
- Oxygen isotopes ($Z = 8$): $\{{}^{14}_{8}\mathrm{O}, {}^{16}_{8}\mathrm{O}\}$.

2. Isobars (Identical A):

- $A = 14$: $\{{}^{14}_{6}\mathrm{C}, {}^{14}_{7}\mathrm{N}, {}^{14}_{8}\mathrm{O}\}$.
- $A = 16$: $\{{}^{16}_{7}\mathrm{N}, {}^{16}_{8}\mathrm{O}\}$.

3. Isotones (Identical N):

- $N = 6$: $\{{}^{12}_{6}\mathrm{C}, {}^{14}_{8}\mathrm{O}\}$.
- $N = 7$: $\{{}^{13}_{6}\mathrm{C}, {}^{14}_{7}\mathrm{N}\}$.
- $N = 8$: $\{{}^{14}_{6}\mathrm{C}, {}^{15}_{7}\mathrm{N}, {}^{16}_{8}\mathrm{O}\}$.

4. Mirror Nuclei ($Z_1 = N_2$ and $N_1 = Z_2$): Consider ${}^{13}_{6}\mathrm{C}$ ($Z = 6, N = 7$) and ${}^{13}_{7}\mathrm{N}$ ($Z = 7, N = 6$). In our set, ${}^{14}_{6}\mathrm{C}$ ($Z = 6, N = 8$) and ${}^{14}_{8}\mathrm{O}$ ($Z = 8, N = 6$) have swapped proton/neutron counts:

$$({}^{14}_{6}\mathrm{C}, {}^{14}_{8}\mathrm{O}) \quad \text{form an authentic mirror isobaric pair!}$$

Exercise 2.5 · The Unified Atomic Mass Unit and Fundamental Mass Conversions ★ Foundational

The unified atomic mass unit (1 u) is defined as exactly 1/12 of the rest mass of an unbound neutral Carbon-12 atom in its ground state.

1. Using Avogadro's constant $N_A = 6.022\,140\,76 \times 10^{23}\ \text{mol}^{-1}$, prove rigorously that $1\ \text{u} = \dfrac{10^{-3}\ \text{kg}}{N_A} \approx 1.660\,539 \times 10^{-27}\ \text{kg}$.
2. The atomic mass of a gold atom ($^{197}_{79}\text{Au}$) is $M = 196.966\,569$ u. Calculate the mass of one single gold atom in:
   (a) Kilograms (kg).
   (b) Grams (g).
3. How many gold atoms are contained in a pure 1.000 kg gold bullion bar?

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.5

1. Rigorous Derivation of 1 u**:** By definition of the mole, exactly 1 mol of Carbon-12 has a mass of exactly $12.000\ \text{g} = 12 \times 10^{-3}\ \text{kg}$. Since 1 mol contains N_A atoms:

$$m(^{12}\text{C}) = \frac{12 \times 10^{-3}\ \text{kg}}{N_A}.$$

By definition of the unified atomic mass unit:

$$1\ \text{u} = \frac{1}{12} m(^{12}\text{C}) = \frac{1}{12}\left(\frac{12 \times 10^{-3}\ \text{kg}}{N_A}\right) = \frac{\mathbf{10^{-3}}\ \text{kg}}{\mathbf{N_A}}.$$

Plugging in $N_A = 6.022\,140\,76 \times 10^{23}\ \text{mol}^{-1}$:

$$1\ \text{u} = \frac{10^{-3}\ \text{kg}}{6.022\,140\,76 \times 10^{23}} = \mathbf{1.660\,539 \times 10^{-27}}\ \text{kg}. \quad \text{Q.E.D.}$$

2. Rest Mass of One Gold Atom:

1. In kilograms:

$$m_{\text{Au}} = 196.966\,569 \times 1.660\,539 \times 10^{-27}\ \text{kg} = \mathbf{3.2707 \times 10^{-25}}\ \text{kg}.$$

2. In grams:

$$m_{\text{Au}} = 3.2707 \times 10^{-22}\ \text{g}.$$

3. Number of Atoms in 1.000 kg Bullion:

$$N_{\text{atoms}} = \frac{m_{\text{bar}}}{m_{\text{Au}}} = \frac{1.000 \text{ kg}}{3.2707 \times 10^{-25} \text{ kg}} = \mathbf{3.057 \times 10^{24}} \text{ atoms.}$$

Alternatively: $n = \frac{1000 \text{ g}}{196.967 \text{ g·mol}^{-1}} = 5.077 \text{ mol} \implies N = 5.077 \times 6.022 \times 10^{23} = \mathbf{3.057 \times 10^{24}}$ atoms.

Exercise 2.6 · Einstein Energy Equivalence of the Atomic Mass Unit ★ Foundational

Using Einstein's mass-energy equivalence relation $E = mc^2$ with speed of light $c = 2.997\,924\,58 \times 10^8$ m · s^{-1} and elementary charge $e = 1.602\,176\,634 \times 10^{-19}$ C:

1. Calculate the energy equivalent of 1 u in Joules (J).
2. Convert this energy into electronvolts (eV) and mega-electronvolts (MeV).
3. State the standard CODATA conversion factor 1 u · c^2 in MeV used throughout nuclear physics.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.6

1. Energy Equivalent in Joules: Using $m = 1$ u $= 1.660\,539\,066 \times 10^{-27}$ kg:

$$\begin{aligned} E = m \cdot c^2 &= (1.660\,539\,066 \times 10^{-27} \text{ kg}) \times (2.997\,924\,58 \times 10^8 \text{ m} \cdot \text{s}^{-1})^2 \\ &= (1.660\,539\,066 \times 10^{-27}) \times (8.987\,551\,787 \times 10^{16}) \\ &= \mathbf{1.492\,418 \times 10^{-10}} \text{ J.} \end{aligned}$$

2. Conversion into eV and MeV: Since 1 eV $= 1.602\,176\,634 \times 10^{-19}$ J:

$$E = \frac{1.492\,418 \times 10^{-10} \text{ J}}{1.602\,176\,634 \times 10^{-19} \text{ J} \cdot \text{eV}^{-1}} = 931\,494\,102 \text{ eV.}$$

Dividing by 10^6 eV · MeV^{-1}:

$$E = \mathbf{931.494} \text{ MeV.}$$

3. Standard Nuclear Conversion Factor:

$$\mathbf{1} \text{ u} \equiv \mathbf{931.494} \text{ MeV}/\mathbf{c^2} \iff \mathbf{E = \Delta m} \text{ (u)} \times \mathbf{931.494} \text{ MeV.}$$

This universal factor allows direct calculation of binding energies without tedious conversions through SI units.

Exercise 2.7 · Average Atomic Mass of Chlorine ★ Foundational

Naturally occurring chlorine consists of two stable isotopes with the following mass spectrometry data:

Isotope	Isotopic Mass (u)	Natural Abundance (%)
$^{35}_{17}\mathrm{Cl}$	34.968 853	75.76%
$^{37}_{17}\mathrm{Cl}$	36.965 903	24.24%

1. Calculate the standard average atomic mass $\bar{M}(\mathrm{Cl})$ of elemental chlorine to four decimal places.
2. Explain why the mass numbers $A_1 = 35$ and $A_2 = 37$ are integers, but the average atomic mass of chlorine reported on the periodic table is ≈ 35.45 u.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.7

1. Average Atomic Mass Calculation: The fractional abundances are $x_1 = 0.7576$ and $x_2 = 0.2424$ (note $x_1 + x_2 = 1.0000$):

$$\begin{aligned}\bar{M}(\mathrm{Cl}) &= x_1 M_1 + x_2 M_2 \\ &= (0.7576 \times 34.968\,853\ \mathrm{u}) + (0.2424 \times 36.965\,903\ \mathrm{u}) \\ &= 26.492\,403\ \mathrm{u} + 8.960\,535\ \mathrm{u} \\ &= \mathbf{35.4529}\ \mathrm{u} \approx \mathbf{35.45}\ \mathrm{u}.\end{aligned}$$

2. Physical Explanation:

- The mass number A is by definition the count of nucleons ($p^+ + n^0$) in a specific nucleus, which is strictly an **integer**.
- The atomic mass of an individual nuclide deviates slightly from A due to nuclear binding energy ($E = mc^2$) and the proton-neutron mass difference.
- The periodic table value represents the **weighted average** of the naturally occurring isotopic mixture. Because terrestrial chlorine is an approximate 3 : 1 blend of $^{35}\mathrm{Cl}$ and $^{37}\mathrm{Cl}$, its average atomic mass is roughly $(3 \times 35 + 1 \times 37)/4 = 35.5$ u, not an integer.

Exercise 2.8 · Determination of Boron Isotopic Abundances ★★ Intermediate

Naturally occurring boron consists exclusively of two stable isotopes: Boron-10 ($^{10}\mathrm{B}$) with isotopic mass $M_1 = 10.0129$ u, and Boron-11 ($^{11}\mathrm{B}$) with isotopic mass $M_2 = 11.0093$ u. The official IUPAC standard atomic weight of natural boron is $\bar{M} = 10.8110$ u.

1. Set up the algebraic system linking the fractional abundances x_1 and x_2 to the average mass.
2. Solve the system to determine the percentage abundances of ^{10}B and ^{11}B to two decimal places.
3. If a nuclear reactor control rod contains $m = 250.0$ g of natural boron, what mass of pure neutron-absorbing ^{10}B is present?

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.8

1. Algebraic Formulation: Let x_1 be the fraction of ^{10}B and x_2 the fraction of ^{11}B:

$$x_1 + x_2 = 1 \implies x_2 = 1 - x_1 \tag{2.1}$$
$$x_1 M_1 + x_2 M_2 = \bar{M} \tag{2.2}$$

2. Calculation of Abundances: Substitute (2.1) into (2.2):

$$x_1 M_1 + (1 - x_1) M_2 = \bar{M} \implies x_1 (M_1 - M_2) + M_2 = \bar{M}$$

$$x_1 = \frac{M_2 - \bar{M}}{M_2 - M_1} = \frac{11.0093 - 10.8110}{11.0093 - 10.0129} = \frac{0.1983}{0.9964} = 0.199016.$$

Thus:

$$x_1(^{10}\text{B}) = 0.1990 \implies \mathbf{19.90}\%,$$
$$x_2(^{11}\text{B}) = 1 - 0.1990 = 0.8010 \implies \mathbf{80.10}\%.$$

3. Mass of ^{10}B in Control Rod: The mole fraction of ^{10}B is $x_1 = 0.1990$. Total moles of boron in 250.0 g:

$$n_{\text{total}} = \frac{m_{\text{rod}}}{\bar{M}} = \frac{250.0 \text{ g}}{10.8110 \text{ g} \cdot \text{mol}^{-1}} = 23.1246 \text{ mol}.$$

Moles of ^{10}B:

$$n(^{10}\text{B}) = x_1 \cdot n_{\text{total}} = 0.199016 \times 23.1246 \text{ mol} = 4.6022 \text{ mol}.$$

Mass of ^{10}B:

$$m(^{10}\text{B}) = n(^{10}\text{B}) \cdot M_1 = 4.6022 \text{ mol} \times 10.0129 \text{ g} \cdot \text{mol}^{-1} = \mathbf{46.08} \text{ g}.$$

Exercise 2.9 · Mass Spectrometry Peak Ratio Analysis of Copper ★★ Intermediate

In an Aston mass spectrometer, a beam of copper ions (Cu^+) is separated into two isotopic parabolic trajectories corresponding to ^{63}Cu ($M_1 = 62.9296$ u) and ^{65}Cu ($M_2 = 64.9278$ u). The ratio of collector ion currents is measured to be:

$$R = \frac{I(^{63}\text{Cu})}{I(^{65}\text{Cu})} = 2.2464.$$

1. Express the fractional abundances x_1 and x_2 in terms of the peak ratio R.
2. Compute the percentage natural abundance of both isotopes.
3. Calculate the average atomic mass of copper based on these spectrometric measurements.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.9

1. Abundances in Terms of Peak Ratio R: Since the ion detector response is linearly proportional to isotopic abundance:

$$\frac{x_1}{x_2} = R \implies x_1 = R \cdot x_2.$$

Substituting into $x_1 + x_2 = 1$:

$$R \cdot x_2 + x_2 = 1 \implies x_2(1+R) = 1 \implies \mathbf{x_2 = \frac{1}{1+R}}, \qquad \mathbf{x_1 = \frac{R}{1+R}}.$$

2. Percentage Abundance Calculation: Given $R = 2.2464$:

$$x_2(^{65}\text{Cu}) = \frac{1}{1+2.2464} = \frac{1}{3.2464} = 0.30803 \implies \mathbf{30.80\%}.$$

$$x_1(^{63}\text{Cu}) = 1 - 0.30803 = 0.69197 \implies \mathbf{69.20\%}.$$

3. Average Atomic Mass of Copper:

$$\begin{aligned} \bar{M}(\text{Cu}) &= x_1 M_1 + x_2 M_2 \\ &= (0.69197 \times 62.9296 \text{ u}) + (0.30803 \times 64.9278 \text{ u}) \\ &= 43.5454 \text{ u} + 19.9997 \text{ u} = \mathbf{63.545} \text{ u}. \end{aligned}$$

This matches the IUPAC standard periodic table atomic weight of copper (63.546 u).

Exercise 2.10 · Three-Isotope System of Magnesium ★★ Intermediate

Naturally occurring magnesium is composed of three stable isotopes:

- $^{24}_{12}$Mg: $M_1 = 23.9850$ u
- $^{25}_{12}$Mg: $M_2 = 24.9858$ u
- $^{26}_{12}$Mg: $M_3 = 25.9826$ u

High-resolution mass spectrometry shows that the ratio of abundances between the two heavier isotopes is $x_3/x_2 = 1.1082$. The standard atomic mass of natural magnesium is $\bar{M} = 24.3050$ u.

1. Set up the system of three equations in three unknowns (x_1, x_2, x_3).
2. Solve the system to determine the exact natural abundance percentages of all three isotopes.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.10

1. System of Equations:

$$x_1 + x_2 + x_3 = 1 \tag{2.3}$$
$$x_1 M_1 + x_2 M_2 + x_3 M_3 = \bar{M} = 24.3050 \tag{2.4}$$
$$x_3 = 1.1082\, x_2 \tag{2.5}$$

2. Solution of the System: Substitute (2.5) into (2.3):

$$x_1 + x_2 + 1.1082\, x_2 = 1 \implies x_1 + 2.1082\, x_2 = 1 \implies x_1 = 1 - 2.1082\, x_2.$$

Now substitute x_1 and x_3 into the mass equation (2.4):

$$(1 - 2.1082\, x_2) M_1 + x_2 M_2 + (1.1082\, x_2) M_3 = \bar{M}$$

$$M_1 + x_2 \left[M_2 + 1.1082\, M_3 - 2.1082\, M_1 \right] = \bar{M}.$$

Compute the bracketed coefficient:

$$\begin{aligned} M_2 + 1.1082\, M_3 - 2.1082\, M_1 &= 24.9858 + 1.1082(25.9826) - 2.1082(23.9850) \\ &= 24.9858 + 28.7939 - 50.5652 = 3.2145 \text{ u}. \end{aligned}$$

The numerator is:

$$\bar{M} - M_1 = 24.3050 - 23.9850 = 0.3200 \text{ u}.$$

Thus:

$$x_2 = \frac{0.3200}{3.2145} = 0.09955 \implies \mathbf{x_2}(^{25}\text{Mg}) = \mathbf{9.96}\%.$$

From (2.5):

$$x_3 = 1.1082 \times 0.09955 = 0.11032 \implies \mathbf{x_3}(^{26}\mathrm{Mg}) = \mathbf{11.03}\%.$$

From (2.3):

$$x_1 = 1 - (0.09955 + 0.11032) = 1 - 0.20987 = 0.79013 \implies \mathbf{x_1}(^{24}\mathrm{Mg}) = \mathbf{79.01}\%.$$

Summary of Natural Abundances:

$$^{24}\mathrm{Mg} : \mathbf{79.01}\%, \qquad ^{25}\mathrm{Mg} : \mathbf{9.96}\%, \qquad ^{26}\mathrm{Mg} : \mathbf{11.03}\%.$$

Exercise 2.11 · Nuclear Mass Defect and Binding of the Deuteron ★★ Intermediate

A deuteron ($^2_1$H or D nucleus) consists of one bound proton and one bound neutron.

- Proton rest mass: $m_p = 1.007\,276$ u
- Neutron rest mass: $m_n = 1.008\,665$ u
- Deuteron nuclear mass: $m_d = 2.013\,553$ u

1. Compute the nuclear mass defect Δm in atomic mass units (u) and in kilograms (kg).
2. Calculate the total binding energy E_L of the deuteron in mega-electronvolts (MeV) and Joules (J).
3. Determine the binding energy per nucleon (E_L/A).

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.11

1. Mass Defect Δm: The constituent particles are 1 p^+ and 1 n^0:

$$m_{\text{constituents}} = m_p + m_n = 1.007\,276 + 1.008\,665 = 2.015\,941 \text{ u}.$$

$$\Delta m = m_{\text{constituents}} - m_d = 2.015\,941 - 2.013\,553 = \mathbf{0.002\,388} \text{ u}.$$

In kilograms:

$$\Delta m = 0.002\,388 \times 1.660\,539 \times 10^{-27} \text{ kg} = \mathbf{3.9654 \times 10^{-30}} \text{ kg}.$$

2. Total Binding Energy E_L: In MeV:

$$E_L = \Delta m \times 931.494 \text{ MeV/u} = 0.002\,388 \text{ u} \times 931.494 \text{ MeV/u} = \mathbf{2.224} \text{ MeV}.$$

In Joules:

$$E_L = 2.2244 \text{ MeV} \times 1.602\,177 \times 10^{-13} \text{ J/MeV} = \mathbf{3.564 \times 10^{-13}} \text{ J}.$$

(Check: $\Delta m \cdot c^2 = 3.9654 \times 10^{-30} \times (2.9979 \times 10^8)^2 = 3.564 \times 10^{-13}$ J).

3. Binding Energy Per Nucleon (E_L/A): For deuterium, $A = 2$:

$$\frac{E_L}{A} = \frac{2.2244 \text{ MeV}}{2} = \mathbf{1.112} \text{ MeV/nucleon}.$$

This low binding energy per nucleon indicates that the deuteron is a relatively loosely bound nucleus.

Exercise 2.12 · Alpha Particle Nuclear Binding and Tightly Bound Systems ★★ Intermediate

An alpha particle is the bare nucleus of Helium-4 ($^4_2\text{He}^{2+}$), containing 2 protons and 2 neutrons.

- Proton mass: $m_p = 1.007\,276$ u
- Neutron mass: $m_n = 1.008\,665$ u
- Helium-4 nuclear mass: $m_\alpha = 4.001\,506$ u

1. Calculate the nuclear mass defect Δm of the alpha particle in u.
2. Calculate the total nuclear binding energy E_L in MeV.
3. Determine the specific binding energy E_L/A.
4. Compare this specific binding energy to that of the deuteron from Exercise 2.11 (1.112 MeV/nucleon). What explains the extraordinary stability of the alpha particle?

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.12

1. Nuclear Mass Defect Δm: Constituents mass ($2\ p^+ + 2\ n^0$):

$$m_{\text{free}} = 2m_p + 2m_n = 2(1.007\,276) + 2(1.008\,665) = 2.014\,552 + 2.017\,330 = 4.031\,882 \text{ u}.$$

$$\Delta m = m_{\text{free}} - m_\alpha = 4.031\,882 - 4.001\,506 = \mathbf{0.030\,376} \text{ u}.$$

2. Total Binding Energy E_L:

$$E_L = \Delta m \times 931.494 \text{ MeV/u} = 0.030\,376 \times 931.494 = \mathbf{28.295} \text{ MeV}.$$

3. Specific Binding Energy (E_L/A): For ^4_2He, $A = 4$:

$$\frac{E_L}{A} = \frac{28.295 \text{ MeV}}{4} = \mathbf{7.074} \text{ MeV/nucleon}.$$

4. Stability Analysis: The binding energy per nucleon of ^{4}He (7.074 MeV/nucleon) is more than **six times larger** than that of deuterium (1.112 MeV/nucleon). Helium-4 is a **doubly**

magic nucleus ($Z = 2, N = 2$), where both protons and neutrons completely fill their lowest nuclear energy shell ($1s_{1/2}$). This exceptionally strong binding explains why alpha particles are emitted as discrete, highly stable entities in radioactive decay chains.

Exercise 2.13 · Binding Energy Calculation Using Atomic Masses ★★ Intermediate

Calculate the nuclear binding energy of Carbon-12 ($^{12}_{6}\mathrm{C}$) using neutral atomic masses.

- Neutral Hydrogen atom rest mass: $m(^{1}_{1}\mathrm{H}) = 1.007\,825$ u
- Neutron rest mass: $m_n = 1.008\,665$ u
- Neutral Carbon-12 atom rest mass: $M(^{12}_{6}\mathrm{C}) \equiv 12.000\,000$ u (by definition)

1. Show why the formula $\Delta m = [Z \cdot m(^{1}\mathrm{H}) + (A - Z) \cdot m_n] - M(^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X})$ automatically accounts for electron masses.
2. Compute Δm for Carbon-12 in u.
3. Calculate the total binding energy E_L in MeV and the binding energy per nucleon E_L/A.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.13

1. Cancellation of Electron Masses: The neutral atom consists of the nucleus plus Z orbital electrons:

$$M(^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X}) \approx m_{\text{nucleus}} + Z \cdot m_e.$$

Similarly, a neutral hydrogen atom is a proton plus one electron:

$$m(^{1}\mathrm{H}) = m_p + m_e.$$

Substituting into the formula:

$$\begin{aligned}\Delta m &= [Z(m_p + m_e) + (A - Z)m_n] - [m_{\text{nucleus}} + Zm_e] \\ &= [Zm_p + Zm_e + (A - Z)m_n] - m_{\text{nucleus}} - Zm_e \\ &= [Zm_p + (A - Z)m_n] - m_{\text{nucleus}}.\end{aligned}$$

The Z electron masses cancel out perfectly! (Electronic binding energies of a few tens of eV are negligible compared to MeV nuclear energies).

2. Mass Defect of $^{12}_{6}\mathrm{C}$: For Carbon-12, $Z = 6$, $N = 6$:

$$\begin{aligned}\Delta m &= 6 \cdot m(^{1}\mathrm{H}) + 6 \cdot m_n - 12.000\,000 \\ &= 6(1.007\,825) + 6(1.008\,665) - 12.000\,000 \\ &= 6.046\,950 + 6.051\,990 - 12.000\,000 \\ &= 12.098\,940 - 12.000\,000 = \mathbf{0.098\,940}\ \text{u}.\end{aligned}$$

3. Total and Specific Binding Energy:

$$E_L = \Delta m \times 931.494 \text{ MeV/u} = 0.098\,940 \times 931.494 = \mathbf{92.162} \text{ MeV}.$$

$$\frac{E_L}{A} = \frac{92.162 \text{ MeV}}{12} = \mathbf{7.680} \text{ MeV/nucleon}.$$

Exercise 2.14 · The Peak of the Aston Stability Curve: Iron-56 ★★ Intermediate

Iron-56 ($^{56}_{26}$Fe) sits at the absolute summit of nuclear stability on the Aston curve.

- Neutral Hydrogen atom: $m(^1_1\text{H}) = 1.007\,825$ u
- Neutron: $m_n = 1.008\,665$ u
- Atomic mass of Iron-56: $M(^{56}_{26}\text{Fe}) = 55.934\,937$ u

1. Calculate the total mass defect Δm of Iron-56.
2. Compute the total nuclear binding energy E_L in MeV.
3. Determine the specific binding energy E_L/A in MeV/nucleon.
4. Explain why iron represents the ultimate thermodynamic "sink" of stellar nucleosynthesis in massive stars.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.14

1. Mass Defect Δm: For $^{56}_{26}$Fe, $Z = 26$ and $N = 56 - 26 = 30$:

$$\begin{aligned} m_{\text{free}} &= 26 \cdot m(^1\text{H}) + 30 \cdot m_n \\ &= 26(1.007\,825) + 30(1.008\,665) \\ &= 26.203\,450 + 30.259\,950 = 56.463\,400 \text{ u}. \end{aligned}$$

$$\Delta m = 56.463\,400 - 55.934\,937 = \mathbf{0.528\,463} \text{ u}.$$

2. Total Binding Energy E_L:

$$E_L = 0.528\,463 \text{ u} \times 931.494 \text{ MeV/u} = \mathbf{492.26} \text{ MeV}.$$

3. Specific Binding Energy (E_L/A):

$$\frac{E_L}{A} = \frac{492.26 \text{ MeV}}{56} = \mathbf{8.790} \text{ MeV/nucleon}.$$

4. Astrophysical Significance: At 8.79 MeV/nucleon, Iron-56 nucleons reside in the deepest nuclear potential well in nature.

- Fusing nuclei lighter than iron releases energy ($\Delta(E_L/A) > 0$).
- Fissioning nuclei heavier than iron releases energy ($\Delta(E_L/A) > 0$).
- However, attempting to fuse iron nuclei or fission iron requires an input of external energy (endothermic). Thus, when a massive star builds an iron core, fusion ceases; unable to support itself against gravity, the core collapses catastrophically, triggering a Type II Supernova.

Exercise 2.15 · Heavy Nuclide Binding: Uranium-238 ★★ Intermediate

Calculate the nuclear binding parameters of Uranium-238 ($^{238}_{92}\mathrm{U}$), the principal isotope of natural uranium:

- $m(^{1}_{1}\mathrm{H}) = 1.007\,825$ u
- $m_n = 1.008\,665$ u
- $M(^{238}_{92}\mathrm{U}) = 238.050\,788$ u

1. Calculate the mass defect Δm in u and in kg.
2. Determine the total binding energy E_L in MeV.
3. Compute the binding energy per nucleon (E_L/A).
4. Why does E_L/A decrease from 8.79 MeV/nucleon (in iron) to 7.57 MeV/nucleon (in uranium)?

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.15

1. Mass Defect Δm: For $^{238}_{92}\mathrm{U}$, $Z = 92$ and $N = 238 - 92 = 146$:

$$\begin{aligned} m_{\text{free}} &= 92(1.007\,825) + 146(1.008\,665) \\ &= 92.719\,900 + 147.265\,090 = 239.984\,990 \text{ u}. \end{aligned}$$

$$\Delta m = 239.984\,990 - 238.050\,788 = \mathbf{1.934\,202} \text{ u}.$$

In kilograms:

$$\Delta m = 1.934\,202 \times 1.660\,539 \times 10^{-27} \text{ kg} = \mathbf{3.2118 \times 10^{-27}} \text{ kg}.$$

2. Total Binding Energy E_L:

$$E_L = 1.934\,202 \text{ u} \times 931.494 \text{ MeV/u} = \mathbf{1801.70} \text{ MeV}.$$

3. Specific Binding Energy (E_L/A):

$$\frac{E_L}{A} = \frac{1801.70 \text{ MeV}}{238} = \mathbf{7.570} \text{ MeV/nucleon}.$$

4. Physical Reason for Decrease: In heavy nuclei ($Z = 92$), long-range Coulomb electrostatic repulsion between protons scales as $Z(Z-1) \sim Z^2$, whereas short-range strong nuclear attractive forces act only between nearest neighbors and scale roughly as A. As Z grows, electrostatic repulsion destabilizes the nucleus, causing E_L/A to drop from 8.79 MeV/nucleon to 7.57 MeV/nucleon.

Exercise 2.16 · Thermodynamic Proof of Nuclear Fusion and Fission ★★★ Advanced

Using the Aston curve binding energies per nucleon calculated in previous exercises:

$$\left(\frac{E_L}{A}\right)_{^2\mathrm{H}} = 1.11\ \mathrm{MeV}, \quad \left(\frac{E_L}{A}\right)_{^4\mathrm{He}} = 7.07\ \mathrm{MeV}, \quad \left(\frac{E_L}{A}\right)_{^{56}\mathrm{Fe}} = 8.79\ \mathrm{MeV}, \quad \left(\frac{E_L}{A}\right)_{^{238}\mathrm{U}} = 7.57\ \mathrm{MeV}.$$

1. Prove mathematically that the hypothetical fusion of two deuterons into an alpha particle:

$$^2_1\mathrm{H} + {}^2_1\mathrm{H} \longrightarrow {}^4_2\mathrm{He}$$

   is overwhelmingly exothermic, and compute the energy released ΔE in MeV.

2. Prove mathematically that the hypothetical symmetric fission of a heavy nucleus ($A \approx 240, E_L/A \approx 7.6$ MeV) into two intermediate fragments ($A \approx 120, E_L/A \approx 8.5$ MeV) is exothermic, and compute the energy released.

3. State the universal thermodynamic criterion for a nuclear transformation to release net energy.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.16

1. Deuteron Fusion Energetics: The total binding energy of reactants:

$$E_{L,\mathrm{reactants}} = 2 \times E_L(^2\mathrm{H}) = 2 \times (2 \times 1.112\ \mathrm{MeV}) = 4.45\ \mathrm{MeV}.$$

The total binding energy of products:

$$E_{L,\mathrm{products}} = E_L(^4\mathrm{He}) = 4 \times 7.074\ \mathrm{MeV} = 28.30\ \mathrm{MeV}.$$

The energy released is the increase in binding energy:

$$Q = E_{L,\mathrm{products}} - E_{L,\mathrm{reactants}} = 28.30 - 4.45 = +\mathbf{23.85}\ \mathrm{MeV}.$$

The reaction is intensely exothermic!

2. Symmetric Fission Energetics: Initial binding energy of heavy parent nucleus ($A = 240$):

$$E_{L,\mathrm{initial}} = 240 \times 7.60\ \mathrm{MeV} = 1824.0\ \mathrm{MeV}.$$

Final binding energy of two intermediate fragments ($A = 120$ each):

$$E_{L,\text{final}} = 2 \times (120 \times 8.50 \text{ MeV}) = 240 \times 8.50 \text{ MeV} = 2040.0 \text{ MeV}.$$

Energy released:

$$Q = E_{L,\text{final}} - E_{L,\text{initial}} = 2040.0 - 1824.0 = +\mathbf{216.0} \text{ MeV}.$$

Over 200 MeV of nuclear energy is liberated per fission event!
3. Universal Thermodynamic Criterion: A nuclear process is spontaneously exoenergetic ($Q > 0$) if and only if:

$$\sum E_{L,\text{products}} > \sum E_{L,\text{reactants}} \iff \left(\frac{E_L}{A}\right)_{\text{products}} > \left(\frac{E_L}{A}\right)_{\text{reactants}}.$$

Any transformation moving nucleons closer to the stability peak at Iron-56 liberates binding energy!

Exercise 2.17 · Energetics of Deuterium-Tritium Thermonuclear Fusion ★★★ Advanced

The principal fusion reaction planned for future magnetic confinement fusion reactors (ITER) is:

$$^{2}_{1}\text{H} + ^{3}_{1}\text{H} \longrightarrow ^{4}_{2}\text{He} + ^{1}_{0}\text{n} + \text{Q}$$

Given the accurate neutral atomic rest masses:

- Deuterium: $m(^{2}_{1}\text{H}) = 2.014\,102$ u
- Tritium: $m(^{3}_{1}\text{H}) = 3.016\,049$ u
- Helium-4: $m(^{4}_{2}\text{He}) = 4.002\,603$ u
- Neutron: $m_n = 1.008\,665$ u

1. Calculate the reaction mass change $\Delta m = m_{\text{reactants}} - m_{\text{products}}$ in u.
2. Determine the energy release Q per fusion event in MeV and in Joules (J).
3. Calculate the total energy released by the complete fusion of $m_{\text{fuel}} = 1.000$ kg of an equimolar D-T fuel mixture (1 mol ^{2}H + 1 mol ^{3}H = 5.030 g of mixture).
4. Compare this energy to burning 1.000 kg of octane gasoline ($\Delta H_{\text{comb}} \approx 44.0$ MJ $\cdot$ kg^{-1}).

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.17

1. Mass Change Δm:

$$m_{\text{reactants}} = m(^2\text{H}) + m(^3\text{H}) = 2.014\,102 + 3.016\,049 = 5.030\,151 \text{ u}.$$
$$m_{\text{products}} = m(^4\text{He}) + m_n = 4.002\,603 + 1.008\,665 = 5.011\,268 \text{ u}.$$

$$\Delta m = 5.030\,151 - 5.011\,268 = \mathbf{0.018\,883} \text{ u}.$$

2. Energy Release Q per Fusion Event:

$$Q = 0.018\,883 \text{ u} \times 931.494 \text{ MeV/u} = \mathbf{17.589} \text{ MeV}.$$

In Joules:

$$Q = 17.589 \text{ MeV} \times 1.602\,177 \times 10^{-13} \text{ J/MeV} = \mathbf{2.818 \times 10^{-12}} \text{ J}.$$

3. Energy of 1.000 kg **Equimolar Fuel Mixture:** One pair of reacting (^{2}H + ^{3}H) nuclei has a combined mass of:

$$m_{\text{pair}} = (2.014\,102 + 3.016\,049) \times 1.660\,539 \times 10^{-27} \text{ kg} = 8.3528 \times 10^{-27} \text{ kg}.$$

Number of fusion reactions in 1.000 kg:

$$N_{\text{reactions}} = \frac{1.000 \text{ kg}}{8.3528 \times 10^{-27} \text{ kg}} = 1.1972 \times 10^{26} \text{ reactions}.$$

Total energy released:

$$E_{\text{total}} = 1.1972 \times 10^{26} \times (2.818 \times 10^{-12} \text{ J}) = \mathbf{3.374 \times 10^{14}} \text{ J} \approx \mathbf{337.4} \text{ TJ}.$$

4. Comparison with Gasoline: For 1.000 kg of gasoline: $E_{\text{gas}} = 4.40 \times 10^7$ J.

$$\frac{E_{\text{fusion}}}{E_{\text{gasoline}}} = \frac{3.374 \times 10^{14} \text{ J}}{4.40 \times 10^7 \text{ J}} \approx \mathbf{7.67 \times 10^6}.$$

One kilogram of fusion fuel yields the equivalent energy of **7.67 million kilograms (7670 tonnes)** of premium gasoline!

Exercise 2.18 · Induced Nuclear Fission of Uranium-235 in a Power Reactor ★★★ Advanced

A typical thermal neutron induced fission reaction of Uranium-235 is:

$$^1_0\text{n} + {}^{235}_{92}\text{U} \longrightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3\ {}^1_0\text{n}$$

Accurate isotopic masses:

- $m(^{235}_{92}\text{U}) = 235.043\,929$ u
- $m(^{141}_{56}\text{Ba}) = 140.914\,411$ u
- $m(^{92}_{36}\text{Kr}) = 91.926\,156$ u
- $m_n = 1.008\,665$ u

1. Calculate the mass change Δm and the energy Q liberated per fission in MeV and J.
2. Calculate the total thermal energy released by the complete fission of $m = 1.000$ kg of pure ^{235}U.

3. A nuclear power plant operates at a thermal efficiency of $\eta = 34.0\%$ and generates a continuous electrical output of $P_e = 1000$ MW (1.000×10^9 W). Calculate the daily consumption mass of ^{235}U in kilograms.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.18

1. Fission Energetics: Net reaction equation after canceling one spectator neutron:

$$^{235}_{92}\mathrm{U} \longrightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 2\ {}^{1}_{0}\mathrm{n}.$$

$m_{\text{reactants}} = 235.043\,929$ u.
$m_{\text{products}} = 140.914\,411 + 91.926\,156 + 2(1.008\,665) = 232.840\,567 + 2.017\,330 = 234.857\,897$ u.

$$\Delta m = 235.043\,929 - 234.857\,897 = \mathbf{0.186\,032}\ \text{u}.$$

$$Q = 0.186\,032\ \text{u} \times 931.494\ \text{MeV/u} = \mathbf{173.29}\ \text{MeV}.$$

In Joules:

$$Q = 173.288\ \text{MeV} \times 1.602\,177 \times 10^{-13}\ \text{J/MeV} = \mathbf{2.776 \times 10^{-11}}\ \text{J}.$$

2. Complete Fission of 1.000 kg **of** ^{235}U**:** Number of ^{235}U nuclei in 1.000 kg:

$$N = \frac{1000\ \text{g}}{235.044\ \text{g} \cdot \text{mol}^{-1}} \times 6.022\,141 \times 10^{23}\ \text{mol}^{-1} = 2.5621 \times 10^{24}\ \text{nuclei}.$$

Total thermal energy:

$$E_{\text{thermal}} = 2.5621 \times 10^{24} \times (2.7764 \times 10^{-11}\ \text{J}) = \mathbf{7.113 \times 10^{13}}\ \text{J} \approx \mathbf{71.13}\ \text{TJ}.$$

3. Daily ^{235}U **Consumption in a** 1000 MW_e **Power Plant:** Thermal power required:

$$P_{\text{th}} = \frac{P_e}{\eta} = \frac{1000\ \text{MW}}{0.340} = 2941.2\ \text{MW} = 2.9412 \times 10^9\ \text{J} \cdot \text{s}^{-1}.$$

Thermal energy required in one full day ($t = 24 \times 3600$ s $= 86\,400$ s):

$$E_{\text{day}} = P_{\text{th}} \times 86\,400\ \text{s} = 2.9412 \times 10^9 \times 86\,400 = 2.5412 \times 10^{14}\ \text{J}.$$

Mass of ^{235}U consumed per day:

$$m_{\text{day}} = \frac{E_{\text{day}}}{E_{\text{thermal per kg}}} = \frac{2.5412 \times 10^{14}\ \text{J}}{7.1134 \times 10^{13}\ \text{J} \cdot \text{kg}^{-1}} = \mathbf{3.573}\ \text{kg/day}.$$

A massive 1000 MW gigawatt power station running continuously consumes merely **3.57** kg of uranium-235 each day!

Exercise 2.19 · Fine Corrections: Nuclear Mass vs. Atomic Mass ★★★ Advanced

The exact neutral atomic mass of Lithium-7 is $M(^7_3\mathrm{Li}) = 7.016\,004$ u.

- Electron rest mass: $m_e = 0.000\,548\,580$ u
- Total electronic binding energy of the three electrons: $B_e = 203.5$ eV

1. Derive the exact relation between the neutral atomic mass $M(^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X})$ and the bare nuclear mass m_{nuc}:
$$m_{\mathrm{nuc}} = M(^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X}) - Z \cdot m_e + \frac{B_e}{c^2}.$$
2. Calculate the bare nuclear mass m_{nuc} of the Lithium-7 nucleus in u to 7 decimal places.
3. Determine the percentage error introduced into the nuclear mass if the electronic binding energy term B_e/c^2 is neglected.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.19

1. Exact Nuclear Mass Derivation: The total mass-energy of a neutral atom is the sum of the nuclear mass, the rest mass of Z electrons, minus their atomic binding energy B_e:
$$M(^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X}) \cdot c^2 = m_{\mathrm{nuc}} \cdot c^2 + Z \cdot m_e \cdot c^2 - B_e.$$
Dividing across by c^2 and solving for m_{nuc}:
$$\mathbf{m_{nuc} = M(^{A}_{Z}X) - Z \cdot m_e + \frac{B_e}{c^2}}. \quad \text{Q.E.D.}$$

2. Calculation for Lithium-7 ($Z = 3$): Convert $B_e = 203.5$ eV into atomic mass units:
$$\frac{B_e}{c^2} = \frac{203.5 \text{ eV}}{931.494 \times 10^6 \text{ eV/u}} = 2.185 \times 10^{-7} \text{ u}.$$
Subtracting the rest mass of 3 electrons:
$$3 \cdot m_e = 3 \times 0.000\,548\,580 \text{ u} = 0.001\,645\,740 \text{ u}.$$
Therefore:
$$\begin{aligned} m_{\mathrm{nuc}}(^7\mathrm{Li}) &= 7.016\,004\,000 - 0.001\,645\,740 + 0.000\,000\,218 \\ &= \mathbf{7.014\,358\,478} \text{ u}. \end{aligned}$$

3. Relative Error Analysis: The neglected term is 2.185×10^{-7} u:
$$\text{Relative Error} = \frac{2.185 \times 10^{-7} \text{ u}}{7.014 \text{ u}} \times 100\% = \mathbf{3.1 \times 10^{-6}}\%.$$
This microscopic discrepancy (0.03 ppm) validates the standard practice in nuclear chemistry of neglecting B_e and approximating $m_{\mathrm{nuc}} \approx M(^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X}) - Zm_e$.

Exercise 2.20 · Relativistic Mass Loss: Chemical vs. Nuclear Reactions ★★★ Advanced

According to special relativity, every exothermic reaction releases energy ΔE at the expense of a microscopic loss of rest mass $\Delta m = \Delta E/c^2$.

1. **Chemical Combustion:** Combustion of 1.000 mol of methane (CH_4) releases $\Delta H^\circ = -890.4$ kJ:

$$CH_4(g) + 2\,O_2(g) \longrightarrow CO_2(g) + 2\,H_2O(l), \qquad \Delta E_{\text{chem}} = 890.4 \text{ kJ}.$$

   Calculate the relativistic mass defect Δm_{chem} in kilograms and compare it to the initial mass of reactants ($M_{\text{initial}} = 80.05$ g).

2. **Nuclear Fission:** Fission of 1.000 mol of Uranium-235 (235.04 g) releases $\Delta E_{\text{nuc}} = 1.67 \times 10^{10}$ kJ. Calculate the nuclear mass defect Δm_{nuc} in kilograms and the percentage of mass converted into energy.

3. Compare the two mass defect fractions and state the profound conceptual distinction between chemical and nuclear energy.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 2.20

1. Chemical Combustion of Methane: Using $\Delta m = \Delta E/c^2$ with $c = 2.9979 \times 10^8$ m · s^{-1}:

$$\Delta m_{\text{chem}} = \frac{890.4 \times 10^3 \text{ J}}{(2.9979 \times 10^8 \text{ m} \cdot \text{s}^{-1})^2} = \frac{8.904 \times 10^5}{8.9876 \times 10^{16}} = \mathbf{9.907 \times 10^{-12}} \text{ kg} \approx \mathbf{9.91} \text{ ng}.$$

Initial mass of reactants:

$$m_{\text{initial}} = M(CH_4) + 2M(O_2) = 16.04 + 2(32.00) = 80.04 \text{ g} = 0.08004 \text{ kg}.$$

Fractional mass loss:

$$\frac{\Delta m_{\text{chem}}}{m_{\text{initial}}} = \frac{9.907 \times 10^{-12} \text{ kg}}{0.08004 \text{ kg}} = \mathbf{1.24 \times 10^{-10}} \quad (\mathbf{1.24 \times 10^{-8}}\%).$$

This mass loss (10 picograms) is entirely beyond detection on any analytical balance, which explains why Lavoisier's law of conservation of mass holds to extraordinary accuracy in chemical reactions!

2. Nuclear Fission of Uranium-235:

$$\Delta m_{\text{nuc}} = \frac{1.67 \times 10^{13} \text{ J}}{8.9876 \times 10^{16} \text{ m}^2 \cdot \text{s}^{-2}} = \mathbf{1.858 \times 10^{-4}} \text{ kg} = \mathbf{0.1858} \text{ g}.$$

Initial mass of reactants: $m_{\text{initial}} = 0.23504$ kg. Fractional mass loss:

$$\frac{\Delta m_{\text{nuc}}}{m_{\text{initial}}} = \frac{0.1858 \text{ g}}{235.04 \text{ g}} \times 100\% = \mathbf{0.0790}\% \approx \mathbf{0.08}\%.$$

3. Fundamental Comparison:

$$\frac{(\Delta m/m)_{\text{nuclear}}}{(\Delta m/m)_{\text{chemical}}} = \frac{7.9 \times 10^{-4}}{1.24 \times 10^{-10}} \approx \mathbf{6.4 \times 10^{6}}.$$

Nuclear reactions rearrange nucleons bound by the **strong nuclear force** ($\sim$ MeV scale), whereas chemical reactions rearrange valence electrons bound by **electromagnetic interactions** ($\sim$ eV scale). Nuclear transformations are approximately **six orders of magnitude (one million times)** more energetic per unit mass than chemical transformations!

Chapter 3

Radioactivity and Nuclear Reactions

Essential Chapter Formulas & Rules · Decay Laws, Kinetics, Activity Units & Radiometric Dating

1. **Soddy-Fajans Conservation Laws:** In any nuclear transformation:

$$\sum Z_{\text{reactants}} = \sum Z_{\text{products}}, \qquad \sum A_{\text{reactants}} = \sum A_{\text{products}}.$$

2. **Fundamental Decay Modes:**

- *Alpha (α):* ${}^{A}_{Z}\text{X} \longrightarrow {}^{A-4}_{Z-2}\text{Y} + {}^{4}_{2}\text{He}$.
- *Beta Minus (β^-):* ${}^{A}_{Z}\text{X} \longrightarrow {}^{A}_{Z+1}\text{Y} + {}^{0}_{-1}\text{e} + \bar{\nu}_e \quad (n \to p + e^- + \bar{\nu}_e)$.
- *Beta Plus (β^+):* ${}^{A}_{Z}\text{X} \longrightarrow {}^{A}_{Z-1}\text{Y} + {}^{0}_{+1}\text{e} + \nu_e \quad (p \to n + e^+ + \nu_e)$.
- *Electron Capture (EC):* ${}^{A}_{Z}\text{X} + {}^{0}_{-1}\text{e} \longrightarrow {}^{A}_{Z-1}\text{Y} + \nu_e$.
- *Gamma De-excitation (γ):* ${}^{A}_{Z}\text{X}^{*} \longrightarrow {}^{A}_{Z}\text{X} + \gamma$.

3. **Decay Kinetics & Exponential Law:**

$$\frac{\mathrm{d}N}{\mathrm{d}t} = -\lambda N(t) \implies N(t) = N_0 e^{-\lambda t}, \qquad m(t) = m_0 e^{-\lambda t}.$$

4. **Half-Life ($t_{1/2}$) & Mean Lifetime (τ):**

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693147}{\lambda}, \qquad \tau = \frac{1}{\lambda} = \frac{t_{1/2}}{\ln 2}.$$

After k half-lives ($t = k \cdot t_{1/2}$): $N(t) = \dfrac{N_0}{2^k}$.

5. **Activity (A) & Units:**

$$A(t) = -\frac{\mathrm{d}N}{\mathrm{d}t} = \lambda N(t) = A_0 e^{-\lambda t}.$$

$$1 \text{ Becquerel (Bq)} = 1 \text{ decay} \cdot \text{s}^{-1}, \qquad 1 \text{ Curie (Ci)} = 3.7 \times 10^{10} \text{ Bq}.$$

6. **Radiometric Dating Formulas:**

- *Radiocarbon (^{14}C):* $t = \dfrac{t_{1/2}}{\ln 2} \ln\left(\dfrac{A_0}{A(t)}\right) = \dfrac{5730 \text{ yr}}{\ln 2} \ln\left(\dfrac{A_0}{A(t)}\right)$.
- *Uranium-Lead ($^{238}\text{U} \to ^{206}$ Pb):* $t = \dfrac{1}{\lambda} \ln\left(1 + \dfrac{N(^{206}\text{Pb})}{N(^{238}\text{U})}\right)$.

Exercise 3.1 · Physical Properties and Shielding of Nuclear Radiations ★ Foundational

Summarize the fundamental physical properties of the four primary ionizing radiations:

1. Complete a comparative table specifying for α, β^-, β^+, and γ:

- Particle nature and composition;
- Electric charge (in units of elementary charge e);
- Rest mass (in atomic mass units u);
- Penetrating power and typical shielding material required to stop it.

2. Explain why alpha radiation is benign outside the human body but exceptionally lethal when inhaled or ingested.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.1

1. Comparative Table of Radiations:

Radiation	Nature	Charge	Rest Mass	Shielding Required
Alpha (α)	${}^{4}_{2}\mathrm{He}^{2+}$ nucleus	$+2\,e$	4.0015 u	Sheet of paper, dead skin layer (40 μm)
Beta Minus (β^-)	Fast electron (e^-)	$-1\,e$	0.000549 u	Few mm of aluminum or Plexiglas
Beta Plus (β^+)	Positron (e^+)	$+1\,e$	0.000549 u	Annihilates with e^- into two 511 keV γ photons
Gamma (γ)	High-energy photon	0	0	Several cm of lead (Pb) or dense concrete

2. Internal vs. External Biological Hazard of α-Radiation:

- **Externally:** Alpha particles have very low penetration (< 0.05 mm in tissue). They are completely stopped by the dead stratum corneum layer of human skin and cannot reach living cells.
- **Internally (Inhaled/Ingested):** Alpha particles possess high mass and $+2e$ charge, resulting in an extremely high **linear energy transfer (LET)**. They dump their full kinetic energy ($\approx$ 4–8 MeV) within a tiny cellular volume ($\approx$ 40 μm, the diameter of 2–3 cells), causing severe double-strand DNA breaks, chromosomal aberrations, and cellular death.

Exercise 3.2 · Soddy-Fajans Conservation Laws in Alpha Decay ★ Foundational

Applying the Soddy-Fajans conservation laws of nucleon number A and charge number Z, complete and balance each of the following alpha decay processes:

1. Uranium-238 alpha decay: $^{238}_{92}\mathrm{U} \longrightarrow ? + {}^{4}_{2}\mathrm{He}$
2. Radium-226 alpha decay: $^{226}_{88}\mathrm{Ra} \longrightarrow ? + {}^{4}_{2}\mathrm{He}$
3. Polonium-210 alpha decay: $^{210}_{84}\mathrm{Po} \longrightarrow ? + {}^{4}_{2}\mathrm{He}$
4. Americium-241 alpha decay (smoke detectors): $^{241}_{95}\mathrm{Am} \longrightarrow ? + {}^{4}_{2}\mathrm{He}$

Identify the daughter element in each case using the periodic table.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.2

For alpha decay: $^{A}_{Z}\mathrm{X} \longrightarrow {}^{A-4}_{Z-2}\mathrm{Y} + {}^{4}_{2}\mathrm{He}$.

1. $^{238}_{92}\mathrm{U}$: $A = 238 - 4 = 234$, $Z = 92 - 2 = 90$ (Thorium, Th):

$$^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}.$$

2. $^{226}_{88}\mathrm{Ra}$: $A = 226 - 4 = 222$, $Z = 88 - 2 = 86$ (Radon, Rn):

$$^{226}_{88}\mathrm{Ra} \longrightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}.$$

3. $^{210}_{84}\mathrm{Po}$: $A = 210 - 4 = 206$, $Z = 84 - 2 = 82$ (Lead, Pb):

$$^{210}_{84}\mathrm{Po} \longrightarrow {}^{206}_{82}\mathrm{Pb} + {}^{4}_{2}\mathrm{He}.$$

4. $^{241}_{95}\mathrm{Am}$: $A = 241 - 4 = 237$, $Z = 95 - 2 = 93$ (Neptunium, Np):

$$^{241}_{95}\mathrm{Am} \longrightarrow {}^{237}_{93}\mathrm{Np} + {}^{4}_{2}\mathrm{He}.$$

Exercise 3.3 · Balancing Beta Minus (β^-) and Beta Plus (β^+) Decays ★ Foundational

Write the complete, balanced nuclear reaction for each decay, including the emitted electron or positron and the corresponding antineutrino ($\bar{\nu}_e$) or neutrino (ν_e):

1. Beta minus decay of Carbon-14 ($^{14}_{6}\mathrm{C}$).
2. Beta minus decay of Iodine-131 ($^{131}_{53}\mathrm{I}$).
3. Beta minus decay of Cobalt-60 ($^{60}_{27}\mathrm{Co}$).
4. Beta plus decay (positron emission) of Carbon-11 ($^{11}_{6}\mathrm{C}$).
5. Beta plus decay of Fluorine-18 ($^{18}_{9}\mathrm{F}$) used in PET scanning.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.3

Beta Minus Rules: ${}^{A}_{Z}X \longrightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e + \bar{\nu}_e$ (with neutron conversion $n \to p + e^- + \bar{\nu}_e$).

1. Carbon-14:
$$ {}^{14}_{6}C \longrightarrow {}^{14}_{7}N + {}^{0}_{-1}e + \bar{\nu}_e. $$
2. Iodine-131:
$$ {}^{131}_{53}I \longrightarrow {}^{131}_{54}Xe + {}^{0}_{-1}e + \bar{\nu}_e. $$
3. Cobalt-60:
$$ {}^{60}_{27}Co \longrightarrow {}^{60}_{28}Ni + {}^{0}_{-1}e + \bar{\nu}_e. $$

Beta Plus Rules: ${}^{A}_{Z}X \longrightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}e + \nu_e$ (with proton conversion $p \to n + e^+ + \nu_e$).

4. Carbon-11:
$$ {}^{11}_{6}C \longrightarrow {}^{11}_{5}B + {}^{0}_{+1}e + \nu_e. $$
5. Fluorine-18:
$$ {}^{18}_{9}F \longrightarrow {}^{18}_{8}O + {}^{0}_{+1}e + \nu_e. $$

Exercise 3.4 · Electron Capture and Gamma De-excitation ★ Foundational

Certain proton-rich nuclides decay competitively via orbital electron capture (EC) rather than positron emission:

1. In rocks, Potassium-40 undergoes electron capture by capturing an inner K-shell electron:
$$ {}^{40}_{19}K + {}^{0}_{-1}e \longrightarrow ? + \nu_e. $$
Identify the daughter nuclide.
2. After electron capture, the atomic electron cloud undergoes reorganization. What characteristic electromagnetic radiation is subsequently emitted?
3. The nuclear isomer Technetium-99m (${}^{99m}_{43}Tc$) decays by isomeric gamma transition. Write the decay equation and explain why its mass number A and atomic number Z remain unchanged.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.4

1. Electron Capture of Potassium-40: $A = 40 + 0 = 40,\ Z = 19 + (-1) = 18$ (Argon, Ar):

$$^{40}_{19}\mathrm{K} + {}^{0}_{-1}\mathrm{e} \longrightarrow {}^{40}_{18}\mathrm{Ar} + \nu_e.$$

2. Characteristic Atomic Radiation: When an inner K-shell electron is captured by the nucleus, a vacancy is created in the innermost shell. An outer electron (from the L or M shell) drops down into the vacancy, releasing the energy difference as a **characteristic X-ray photon** (K_α or K_β X-rays) or through the emission of an Auger electron.

3. Isomeric Transition of Technetium-99m:

$$^{99m}_{43}\mathrm{Tc} \longrightarrow {}^{99}_{43}\mathrm{Tc} + \gamma \quad (E_\gamma = 140.5 \text{ keV}).$$

The "m" denotes a metastable nuclear excited state. A gamma decay is a pure electromagnetic transition where the nucleus sheds excess excitation energy without altering the number of protons ($Z = 43$) or neutrons ($N = 56$).

Exercise 3.5 · Historic Nuclear Transmutation and Discovery Reactions ★ Foundational

Write the balanced nuclear equations for three monumental discoveries in modern physics:

1. **Discovery of the Proton (Rutherford, 1919):** Nitrogen-14 bombarded with alpha particles produces an unknown nuclide and a free proton ($^{1}_{1}\mathrm{p}$ or $^{1}_{1}\mathrm{H}$).
2. **Discovery of the Neutron (Chadwick, 1932):** Beryllium-9 bombarded with alpha particles produces Carbon-12 and a neutral particle.
3. **First Artificial Radioactivity (Irène & Frédéric Joliot-Curie, 1934):** Aluminum-27 bombarded with alpha particles ejects a neutron to form a radioactive phosphorus isotope ($^{30}_{15}\mathrm{P}$), which subsequently decays by positron emission. Write both reactions.

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Complete Step-by-Step Solution 3.5

1. Rutherford's Proton Discovery (1919): Reactants: $A = 14 + 4 = 18,\ Z = 7 + 2 = 9$. Known product: ${}^{1}_{1}\mathrm{p}$. Daughter nuclide: $A = 18 - 1 = 17,\ Z = 9 - 1 = 8$ (Oxygen, O):

$$ {}^{14}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \longrightarrow {}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{p} \quad \text{(Notation: } {}^{14}\mathrm{N}(\alpha,\mathrm{p})^{17}\mathrm{O}\text{)}. $$

2. Chadwick's Neutron Discovery (1932): Reactants: $A = 9 + 4 = 13,\ Z = 4 + 2 = 6$. Known product: ${}^{12}_{6}\mathrm{C}$. Neutral particle: $A = 13 - 12 = 1,\ Z = 6 - 6 = 0$ (Neutron, ${}^{1}_{0}\mathrm{n}$):

$$ {}^{9}_{4}\mathrm{Be} + {}^{4}_{2}\mathrm{He} \longrightarrow {}^{12}_{6}\mathrm{C} + {}^{1}_{0}\mathrm{n} \quad \text{(Notation: } {}^{9}\mathrm{Be}(\alpha,\mathrm{n})^{12}\mathrm{C}\text{)}. $$

3. Joliot-Curie Artificial Radioactivity (1934):

- Bombardment synthesis: Reactants: $A = 27 + 4 = 31,\ Z = 13 + 2 = 15$. Ejected particle: ${}^{1}_{0}\mathrm{n}$.

$$ {}^{27}_{13}\mathrm{Al} + {}^{4}_{2}\mathrm{He} \longrightarrow {}^{30}_{15}\mathrm{P} + {}^{1}_{0}\mathrm{n}. $$

- Radioactive decay of synthesized phosphorus-30 ($t_{1/2} = 2.50$ min):

$$ {}^{30}_{15}\mathrm{P} \longrightarrow {}^{30}_{14}\mathrm{Si} + {}^{0}_{+1}\mathrm{e} + \nu_e. $$

Exercise 3.6 · Decay Constant, Half-Life, and Mean Lifetime ★ Foundational

The decay kinetics of a radioactive nuclide are governed by the differential equation $\mathrm{d}N/\mathrm{d}t = -\lambda N$.

1. Integrate this differential equation analytically with initial condition $N(0) = N_0$ to derive $N(t) = N_0 e^{-\lambda t}$.
2. Derive the exact relationship between the half-life $t_{1/2}$ and the decay constant λ.
3. Radon-222 (${}^{222}_{86}\mathrm{Rn}$) has a half-life of $t_{1/2} = 3.823$ days. Calculate its decay constant λ in day^{-1} and in SI units (s^{-1}).
4. Calculate the mean lifetime $\tau = 1/\lambda$ of a Radon-222 nucleus.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.6

1. Analytical Integration: Separate variables:

$$ \frac{\mathrm{d}N}{N} = -\lambda \mathrm{d}t. $$

Integrate from $t = 0$ ($N = N_0$) to time t ($N = N(t)$):

$$\int_{N_0}^{N(t)} \frac{\mathrm{d}N}{N} = -\lambda \int_0^t \mathrm{d}t \implies \ln\left(\frac{N(t)}{N_0}\right) = -\lambda t.$$

Exponentiating both sides:

$$\frac{N(t)}{N_0} = e^{-\lambda t} \implies \mathbf{N(t) = N_0 e^{-\lambda t}}. \quad \text{Q.E.D.}$$

2. Derivation of Half-Life Formula: By definition of the half-life $t_{1/2}$, at $t = t_{1/2}$, exactly half the nuclei remain: $N(t_{1/2}) = N_0/2$.

$$\frac{N_0}{2} = N_0 e^{-\lambda t_{1/2}} \implies e^{\lambda t_{1/2}} = 2 \implies \lambda t_{1/2} = \ln 2.$$

$$\mathbf{t_{1/2}} = \frac{\ln \mathbf{2}}{\lambda} \approx \frac{\mathbf{0.693147}}{\lambda}, \qquad \lambda = \frac{\ln \mathbf{2}}{\mathbf{t_{1/2}}}.$$

3. Calculations for Radon-222:

$$\lambda = \frac{\ln 2}{3.823 \text{ days}} = \mathbf{0.1813} \text{ day}^{-1}.$$

Converting days to seconds (1 day = 86 400 s):

$$t_{1/2} = 3.823 \times 86\,400 \text{ s} = 330\,307.2 \text{ s}.$$

$$\lambda = \frac{\ln 2}{330\,307.2 \text{ s}} = \mathbf{2.0985 \times 10^{-6}} \text{ s}^{-1}.$$

4. Mean Lifetime τ:

$$\tau = \frac{1}{\lambda} = \frac{1}{0.1813 \text{ day}^{-1}} = \mathbf{5.515} \text{ days} \approx \mathbf{4.765 \times 10^5} \text{ s}.$$

Notice that $\tau = t_{1/2}/\ln 2 \approx 1.443\, t_{1/2}$.

Exercise 3.7 · Absolute Activity: The Becquerel and the Curie ★ Foundational

The historical Curie unit (1 Ci) was originally defined by Marie Curie as the activity of exactly 1.000 g of pure Radium-226 ($^{226}_{88}$Ra).

- Molar mass of Radium-226: $M = 226.025 \text{ g} \cdot \text{mol}^{-1}$
- Half-life: $t_{1/2} = 1600$ years (1 yr = 365.25 days = 3.15576×10^7 s)
- $N_A = 6.022\,141 \times 10^{23} \text{ mol}^{-1}$

1. Calculate the decay constant λ of Radium-226 in s^{-1}.
2. Determine the number of radium nuclei N in 1.000 g of ^{226}Ra.
3. Compute the absolute activity $A = \lambda N$ in Becquerels (Bq).
4. Compare your result to the modern exact definition: 1 Ci $\equiv 3.700 \times 10^{10}$ Bq.

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Complete Step-by-Step Solution 3.7

1. Decay Constant λ in s^{-1}: Half-life in seconds:

$$t_{1/2} = 1600 \times 3.15576 \times 10^{7}\ \mathrm{s} = 5.0492 \times 10^{10}\ \mathrm{s}.$$

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{0.693147}{5.0492 \times 10^{10}\ \mathrm{s}} = \mathbf{1.3728 \times 10^{-11}}\ \mathrm{s}^{-1}.$$

2. Number of Radium Nuclei in 1.000 g:

$$N = \frac{m}{M} \cdot N_A = \frac{1.000\ \mathrm{g}}{226.025\ \mathrm{g \cdot mol^{-1}}} \times 6.022\,141 \times 10^{23}\ \mathrm{mol}^{-1} = \mathbf{2.6644 \times 10^{21}}\ \text{nuclei}.$$

3. Absolute Activity A in Becquerels:

$$A = \lambda N = (1.3728 \times 10^{-11}\ \mathrm{s}^{-1}) \times (2.6644 \times 10^{21}) = \mathbf{3.658 \times 10^{10}}\ \mathrm{Bq}.$$

4. Comparison with Standard Definition: The calculated activity of 1 g of ^{226}Ra (3.66×10^{10} Bq) differs by only 1.1% from the standardized definition of the Curie (3.700×10^{10} Bq), which was established internationally in 1953 as an exact rounded reference benchmark.

Exercise 3.8 · Radioactive Mass Decay of Iodine-131 ★★ Intermediate

Iodine-131 ($^{131}_{53}$I) is widely used in hospital nuclear medicine for the therapeutic destruction of hyperactive thyroid tissue. It decays by β^- emission with a half-life of $t_{1/2} = 8.020$ days. A hospital pharmacy receives an initial sample containing $m_0 = 20.00$ mg of pure ^{131}I.

1. Calculate the remaining mass of ^{131}I after:
   (a) Exactly two half-lives ($t = 16.04$ days).
   (b) Exactly three half-lives ($t = 24.06$ days).
   (c) Exactly 30.00 days.
2. What percentage of the initial radioactivity has decayed after 30.00 days?
3. How many days must elapse until only 0.100% (1/1000) of the initial iodine remains?

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Complete Step-by-Step Solution 3.8

1. Remaining Mass:

1. After $k = 2$ half-lives:

$$m(16.04 \text{ d}) = \frac{m_0}{2^2} = \frac{20.00 \text{ mg}}{4} = \mathbf{5.000} \text{ mg}.$$

2. After $k = 3$ half-lives:

$$m(24.06 \text{ d}) = \frac{m_0}{2^3} = \frac{20.00 \text{ mg}}{8} = \mathbf{2.500} \text{ mg}.$$

3. After $t = 30.00$ days: First find λ: $\lambda = \frac{\ln 2}{8.020 \text{ d}} = 0.086427 \text{ day}^{-1}$.

$$m(30.0 \text{ d}) = m_0 e^{-\lambda t} = 20.00 \times e^{-0.086427 \times 30.00} = 20.00 \times e^{-2.5928} = \mathbf{1.496} \text{ mg}.$$

2. Percentage Decayed after 30.00 days**:** Fraction remaining: $\frac{1.496}{20.00} = 0.0748 \implies 7.48\%$.

$$\text{Percentage Decayed} = 100\% - 7.48\% = \mathbf{92.52}\%.$$

3. Time for 0.100% **Remaining** ($N(t)/N_0 = 10^{-3}$)**:**

$$\frac{N(t)}{N_0} = e^{-\lambda t} = 0.00100 \implies -\lambda t = \ln(0.00100) = -6.907755.$$

$$t = \frac{6.907755}{0.086427 \text{ day}^{-1}} = \mathbf{79.93} \text{ days} \approx \mathbf{80} \text{ days}.$$

(Rule of thumb: 10 half-lives reduces activity by $2^{10} = 1024 \approx 1000$; here $10 \times 8.02 = 80.2$ days).

Exercise 3.9 · Determination of Radioactive Half-Life from Activity Data ★★ Intermediate

A laboratory technician measures the count rate (activity in counts per minute, cpm) of a newly isolated beta-emitting radioisotope using a Geiger-Müller counter. The recorded data are:

Time t (hours)	0	3.0	6.0	9.0	12.0	18.0
Activity $A(t)$ (cpm)	8400	5940	4200	2970	2100	1050

1. By inspecting the activity values at $t = 0$ h, $t = 6.0$ h, $t = 12.0$ h, and $t = 18.0$ h, deduce the half-life $t_{1/2}$ directly.
2. Calculate the decay constant λ in h^{-1} and in s^{-1}.

3. Predict the expected activity of the sample at $t = 24.0$ hours and at $t = 48.0$ hours.

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Complete Step-by-Step Solution 3.9

1. Direct Deduction of Half-Life: Observe the sequence of recorded activities:

- At $t = 0$ h: $A_0 = 8400$ cpm.
- At $t = 6.0$ h: $A(6) = 4200$ cpm $= A_0/2$.
- At $t = 12.0$ h: $A(12) = 2100$ cpm $= A_0/4$.
- At $t = 18.0$ h: $A(18) = 1050$ cpm $= A_0/8$.

In every interval of 6.0 hours, the activity is halved precisely. Therefore:

$$\mathbf{t_{1/2} = 6.00} \text{ hours.}$$

(This corresponds to the well-known medical radioisotope Technetium-99m!).

2. Decay Constant λ:

$$\lambda = \frac{\ln 2}{6.00 \text{ h}} = \mathbf{0.1155} \text{ h}^{-1}.$$

In SI units (1 h = 3600 s):

$$\lambda = \frac{0.115525 \text{ h}^{-1}}{3600 \text{ s} \cdot \text{h}^{-1}} = \mathbf{3.209 \times 10^{-5}} \text{ s}^{-1}.$$

3. Predicted Activities:

- At $t = 24.0$ h (4 half-lives):

$$A(24) = \frac{A_0}{2^4} = \frac{8400}{16} = \mathbf{525} \text{ cpm.}$$

- At $t = 48.0$ h (8 half-lives):

$$A(48) = \frac{A_0}{2^8} = \frac{8400}{256} = \mathbf{32.8} \text{ cpm.}$$

Exercise 3.10 · Specific Activity of Pure Phosphorus-32 ★★ Intermediate

Phosphorus-32 ($^{32}_{15}$P, $M = 31.974$ g $\cdot$ mol^{-1}) is a pure β^- emitter with $t_{1/2} = 14.26$ days, used in molecular biology for radiolabeling nucleic acids.

1. Define the **specific activity** (a) of a radioactive source.

2. Derive the algebraic formula for the specific activity a of a carrier-free pure radionuclide in terms of N_A, M, and $t_{1/2}$.

3. Calculate the specific activity of carrier-free pure ^{32}P in:

   (a) Becquerels per gram ($\mathrm{Bq} \cdot \mathrm{g}^{-1}$).

   (b) Curies per gram ($\mathrm{Ci} \cdot \mathrm{g}^{-1}$).

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.10

1. Definition of Specific Activity: The specific activity a is the activity per unit mass of the substance:

$$a = \frac{A}{m} \quad [\mathrm{Bq} \cdot \mathrm{g}^{-1} \text{ or } \mathrm{Ci} \cdot \mathrm{g}^{-1}].$$

2. Algebraic Derivation: For a sample of mass m of a pure radionuclide with molar mass M:

$$N = \frac{m}{M} \cdot N_A.$$

The activity is $A = \lambda N = \frac{\ln 2}{t_{1/2}} \cdot \frac{m}{M} \cdot N_A$. Dividing by m:

$$\mathbf{a = \frac{\ln 2 \cdot N_A}{M \cdot t_{1/2}}}. \quad \text{Q.E.D.}$$

3. Numerical Calculation for Phosphorus-32: Half-life in seconds:

$$t_{1/2} = 14.26 \times 86\,400 \text{ s} = 1.232\,064 \times 10^6 \text{ s}.$$

1. Specific activity in $\mathrm{Bq} \cdot \mathrm{g}^{-1}$:

$$a = \frac{0.693\,147 \times 6.022\,141 \times 10^{23} \text{ mol}^{-1}}{31.974 \text{ g} \cdot \text{mol}^{-1} \times 1.232\,064 \times 10^6 \text{ s}}$$
$$= \frac{4.174\,23 \times 10^{23}}{3.9394 \times 10^7} = \mathbf{1.0596 \times 10^{16}} \text{ Bq} \cdot \text{g}^{-1}.$$

2. Specific activity in $\mathrm{Ci} \cdot \mathrm{g}^{-1}$:

$$a = \frac{1.0596 \times 10^{16} \text{ Bq} \cdot \text{g}^{-1}}{3.700 \times 10^{10} \text{ Bq} \cdot \text{Ci}^{-1}} = \mathbf{286\,380} \text{ Ci} \cdot \text{g}^{-1} \approx \mathbf{2.86 \times 10^5} \text{ Ci} \cdot \text{g}^{-1}.$$

A single milligram of carrier-free ^{32}P carries an astonishing activity of 286 Ci!

Exercise 3.11 · Nuclear Medicine Radiopharmaceutical Dosage Protocol ★★ Intermediate

A nuclear medicine department orders Technetium-99m ($^{99m}_{43}\text{Tc}$, $t_{1/2} = 6.010$ hours) from a regional radiopharmacy. The radiopharmaceutical is calibrated to have an activity of $A_0 = 740.0$ MBq (20.00 mCi) upon delivery at the hospital at 08 : 00 AM.

1. A patient is scheduled for a cardiac perfusion SPECT scan at 14 : 30 (2 : 30 PM) on the same day. Calculate the activity A remaining in the syringe at the time of injection.
2. If the scan protocol prescribes a required injected dose of exactly 400.0 MBq, at what exact time of day should the patient have been injected with this syringe?
3. If unused, how much activity remains in the syringe after 24.0 hours (at 08 : 00 AM the next day)?

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.11

1. Activity at 14 : 30 **(Elapsed time** $\Delta t = 6.50$ hours**):** Decay constant:

$$\lambda = \frac{\ln 2}{6.010\ \text{h}} = 0.115332\ \text{h}^{-1}.$$

Remaining activity:

$$A(6.50\ \text{h}) = A_0 e^{-\lambda t} = 740.0\ \text{MBq} \times e^{-0.115332 \times 6.50} = 740.0 \times e^{-0.74966} = \mathbf{349.5}\ \text{MBq}.$$

2. Injection Time for Exactly 400.0 MBq**:**

$$\frac{A(t)}{A_0} = \frac{400.0}{740.0} = 0.54054 \implies e^{-\lambda t} = 0.54054$$

$$-\lambda t = \ln(0.54054) = -0.61519 \implies t = \frac{0.61519}{0.115332\ \text{h}^{-1}} = 5.334\ \text{hours}.$$

Convert decimal hours to hours and minutes:

$$0.334\ \text{h} \times 60\ \text{min/h} = 20.04\ \text{min} \approx 20\ \text{minutes}.$$

Adding 5 hours 20 minutes to 08 : 00 AM:

$$\text{Injection Time} = \mathbf{13} : \mathbf{20}\ (\mathbf{1} : \mathbf{20}\ \text{PM}).$$

3. Residual Activity After 24.0 Hours ($k = 24.0/6.010 = 3.993 \approx 4$ **half-lives):**

$$A(24\ \text{h}) = 740.0 \times e^{-0.115332 \times 24.0} = 740.0 \times e^{-2.7680} = \mathbf{46.33}\ \text{MBq} \approx \mathbf{1.25}\ \text{mCi}.$$

Exercise 3.12 · Counting Particle Emissions in Natural Decay Series ★★ Intermediate

The natural radioactive decay series cascade through multiple intermediate radionuclides before terminating at a stable lead isotope (Pb, $Z = 82$).

1. **Thorium Series ($4n$):** Thorium-232 ($^{232}_{90}\mathrm{Th}$) decays through a series of α and β^- emissions to stable Lead-208 ($^{208}_{82}\mathrm{Pb}$). Determine the exact number of alpha particles (N_α) and beta particles (N_β) emitted per transformed parent atom.
2. **Actinium Series ($4n{+}3$):** Uranium-235 ($^{235}_{92}\mathrm{U}$) decays to stable Lead-207 ($^{207}_{82}\mathrm{Pb}$). Determine N_α and N_β.
3. State the general algebraic formula for N_α and N_β in any decay chain transitioning from $^{\mathrm{A}_1}_{\mathrm{Z}_1}\mathrm{X}$ to $^{\mathrm{A}_2}_{\mathrm{Z}_2}\mathrm{Y}$.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.12

3. General Algebraic Formula: Let the overall chain transformation be:

$$^{\mathrm{A}_1}_{\mathrm{Z}_1}\mathrm{X} \longrightarrow {}^{\mathrm{A}_2}_{\mathrm{Z}_2}\mathrm{Y} + \mathrm{N}_\alpha\ {}^{4}_{2}\mathrm{He} + \mathrm{N}_\beta\ {}^{0}_{-1}\mathrm{e}.$$

Conservation of mass number A:

$$A_1 = A_2 + 4N_\alpha + 0 \implies \mathbf{N_\alpha = \frac{A_1 - A_2}{4}}.$$

Conservation of charge number Z:

$$Z_1 = Z_2 + 2N_\alpha - N_\beta \implies \mathbf{N_\beta = 2N_\alpha - (Z_1 - Z_2)}.$$

1. Thorium Series ($^{232}_{90}\mathrm{Th} \longrightarrow {}^{208}_{82}\mathrm{Pb}$):

$$N_\alpha = \frac{232 - 208}{4} = \frac{24}{4} = \mathbf{6}\ \alpha\text{-particles}.$$
$$N_\beta = 2(6) - (90 - 82) = 12 - 8 = \mathbf{4}\ \beta^-\text{-particles}.$$

Overall balanced equation:

$$^{232}_{90}\mathrm{Th} \longrightarrow {}^{208}_{82}\mathrm{Pb} + 6\ {}^{4}_{2}\mathrm{He} + 4\ {}^{0}_{-1}\mathrm{e} + 4\ \bar{\nu}_e.$$

2. Actinium Series ($^{235}_{92}\mathrm{U} \longrightarrow {}^{207}_{82}\mathrm{Pb}$):

$$N_\alpha = \frac{235 - 207}{4} = \frac{28}{4} = \mathbf{7}\ \alpha\text{-particles}.$$
$$N_\beta = 2(7) - (92 - 82) = 14 - 10 = \mathbf{4}\ \beta^-\text{-particles}.$$

Overall balanced equation:

$$^{235}_{92}\mathrm{U} \longrightarrow {}^{207}_{82}\mathrm{Pb} + 7\ {}^{4}_{2}\mathrm{He} + 4\ {}^{0}_{-1}\mathrm{e} + 4\ \bar{\nu}_e.$$

Exercise 3.13 · Helium Accumulation from Alpha Emitters ★★ Intermediate

Alpha particles are energetic helium nuclei that capture two stray electrons from surrounding matter to become neutral gaseous Helium-4 ($^4_2\mathrm{He}$). A sealed ampoule contains $m = 1.000$ g of pure Radium-226 chloride ($\mathrm{RaCl_2}$, $M = 296.93\ \mathrm{g \cdot mol^{-1}}$). The radium decays by alpha emission with a half-life of $t_{1/2} = 1600$ years.

1. Compute the number of $^{226}\mathrm{Ra}$ nuclei initially present in the ampoule.
2. Assuming the ampoule remains sealed for exactly $t = 10.0$ years, calculate the number of radium nuclei that decay.
3. Assuming each decaying radium nucleus yields one helium atom, determine the volume of helium gas accumulated inside the ampoule measured at standard temperature and pressure (STP : $V_0 = 22.414\ \mathrm{L \cdot mol^{-1}}$).

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.13

1. Number of Initial Radium Nuclei (N_0):

$$n(\mathrm{RaCl_2}) = \frac{1.000\ \mathrm{g}}{296.93\ \mathrm{g \cdot mol^{-1}}} = 3.3678 \times 10^{-3}\ \mathrm{mol}.$$

Since each formula unit contains one radium atom:

$$N_0 = 3.3678 \times 10^{-3}\ \mathrm{mol} \times 6.022\,141 \times 10^{23}\ \mathrm{mol^{-1}} = \mathbf{2.0281 \times 10^{21}}\ \mathrm{nuclei}.$$

2. Number of Decays in 10.0 years**:** Decay constant in $\mathrm{yr^{-1}}$:

$$\lambda = \frac{\ln 2}{1600\ \mathrm{yr}} = 4.332 \times 10^{-4}\ \mathrm{yr^{-1}}.$$

Since $\lambda t = 4.332 \times 10^{-4} \times 10.0 = 4.332 \times 10^{-3} \ll 1$, we can use the first-order Taylor expansion $1 - e^{-\lambda t} \approx \lambda t$:

$$\Delta N = N_0(1 - e^{-\lambda t}) \approx N_0 \cdot \lambda \cdot t.$$

$$\Delta N = (2.0281 \times 10^{21}) \times (4.332 \times 10^{-3}) = \mathbf{8.786 \times 10^{18}}\ \mathrm{decays}.$$

3. Volume of Helium Gas at STP: Number of moles of helium gas formed:

$$n(\mathrm{He}) = \frac{\Delta N}{N_A} = \frac{8.786 \times 10^{18}}{6.022\,141 \times 10^{23}} = 1.4589 \times 10^{-5}\ \mathrm{mol}.$$

Volume at STP:

$$V(\mathrm{He}) = n \cdot V_0 = 1.4589 \times 10^{-5}\ \mathrm{mol} \times 22.414\ \mathrm{L \cdot mol^{-1}} = 3.270 \times 10^{-4}\ \mathrm{L} = \mathbf{0.327}\ \mathrm{mL}.$$

Over a single decade, one gram of radium chloride generates nearly a third of a milliliter of pure helium gas!

Exercise 3.14 · Secular Equilibrium in the Uranium Decay Series ★★ Intermediate

In natural uranium minerals that have remained undisturbed for geological epochs, parent Uranium-238 (^{238}U, $t_{1/2,1} = 4.468 \times 10^9$ yr) exists in **secular equilibrium** with its immediate daughter Thorium-234 (^{234}Th, $t_{1/2,2} = 24.10$ days).

1. State the condition of secular equilibrium and show why the activities of parent and daughter are equal: $A_1 = A_2$.
2. Derive the equilibrium mass ratio $\dfrac{m_2}{m_1} = \dfrac{t_{1/2,2}}{t_{1/2,1}} \cdot \dfrac{M_2}{M_1}$.
3. Calculate the mass of Thorium-234 present in secular equilibrium with $m_1 = 1.000$ kg of natural Uranium-238.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.14

1. Secular Equilibrium Condition: When the parent has a half-life enormously longer than the daughter ($t_{1/2,1} \gg t_{1/2,2}$, here 4.5×10^9 yr $\gg$ 24 days), the parent's decay rate is virtually constant. The rate of creation of daughter nuclei ($\lambda_1 N_1$) balances the rate of daughter decay ($\lambda_2 N_2$):

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = \lambda_1 N_1 - \lambda_2 N_2 = 0 \implies \boldsymbol{\lambda_1 N_1 = \lambda_2 N_2} \iff \boldsymbol{A_1 = A_2}.$$

2. Derivation of Equilibrium Mass Ratio: From $\lambda_1 N_1 = \lambda_2 N_2$:

$$\frac{N_2}{N_1} = \frac{\lambda_1}{\lambda_2} = \frac{\ln 2/t_{1/2,1}}{\ln 2/t_{1/2,2}} = \frac{t_{1/2,2}}{t_{1/2,1}}.$$

Substitute $N = \frac{m}{M} N_A$:

$$\frac{m_2/M_2}{m_1/M_1} = \frac{t_{1/2,2}}{t_{1/2,1}} \implies \boldsymbol{\frac{m_2}{m_1} = \frac{t_{1/2,2}}{t_{1/2,1}} \cdot \frac{M_2}{M_1}}. \quad \text{Q.E.D.}$$

3. Calculation of Equilibrium Mass of ^{234}Th: Convert half-lives to identical units (days):

$$t_{1/2,1} = 4.468 \times 10^9 \times 365.25 \text{ days} = 1.632 \times 10^{12} \text{ days}.$$

$$m_2 = m_1 \times \left(\frac{24.10 \text{ days}}{1.632 \times 10^{12} \text{ days}}\right) \times \left(\frac{234.04}{238.05}\right) = 1000 \text{ g} \times (1.4767 \times 10^{-11}) \times 0.98315.$$

$$m_2 = \boldsymbol{1.452 \times 10^{-8}} \text{ g} = \boldsymbol{14.52} \text{ ng}.$$

In an entire kilogram of uranium ore, only 14.5 nanograms of thorium-234 exists at any given instant, yet its activity matches that of the entire kilogram of uranium!

Exercise 3.15 · Radiocarbon Archaeological Dating of Ancient Wood ★★ Intermediate

Living plant matter maintains a constant dynamic equilibrium with atmospheric Carbon-14, exhibiting a specific activity of $A_0 = 15.30$ disintegrations per minute per gram of total carbon ($\text{dpm} \cdot \text{g}^{-1}$). After the organism dies, carbon exchange ceases and ^{14}C decays with a half-life of $t_{1/2} = 5730$ years.

1. A wooden beam excavated from the tomb of an Egyptian pharaoh yields a measured specific activity of $A(t) = 9.42\ \text{dpm} \cdot \text{g}^{-1}$.
2. Derive the dating formula expressing age t in terms of $t_{1/2}, A_0$, and $A(t)$.
3. Calculate the age of the wooden beam to the nearest decade.
4. In what calendar era (BCE) was the tree felled?

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.15

1 & 2. Derivation of Radiocarbon Dating Formula: From the decay equation $A(t) = A_0 e^{-\lambda t}$:

$$\frac{A(t)}{A_0} = e^{-\lambda t} \implies \ln\left(\frac{A(t)}{A_0}\right) = -\lambda t \implies \ln\left(\frac{A_0}{A(t)}\right) = \lambda t.$$

Since $\lambda = \dfrac{\ln 2}{t_{1/2}}$:

$$\mathbf{t = \frac{1}{\lambda} \ln\left(\frac{A_0}{A(t)}\right) = \frac{t_{1/2}}{\ln 2} \ln\left(\frac{A_0}{A(t)}\right)}. \quad \text{Q.E.D.}$$

3. Numerical Calculation of Age: Given $A_0 = 15.30\ \text{dpm} \cdot \text{g}^{-1}$, $A(t) = 9.42\ \text{dpm} \cdot \text{g}^{-1}$, and $t_{1/2} = 5730$ yr:

$$\frac{A_0}{A(t)} = \frac{15.30}{9.42} = 1.6242.$$

$$\ln(1.6242) = 0.48501.$$

$$t = \frac{5730\ \text{yr}}{0.693147} \times 0.48501 = 8266.6 \times 0.48501 = \mathbf{4009}\ \text{years} \approx \mathbf{4010}\ \text{years}.$$

4. Calendar Era: Subtracting current date (2026):

$$\text{Date} = 2026 - 4010 = -1984 \implies \approx \mathbf{1980}\ \text{BCE} \quad \text{(Middle Kingdom of Ancient Egypt)}.$$

Exercise 3.16 · Dating with Measurement Uncertainty Propagation ★★★ Advanced

Charcoal from the prehistoric Chauvet Cave (France) is analyzed by accelerator mass spectrometry (AMS). The remaining ^{14}C activity is measured to be $A = 1.85 \pm 0.05\%$ of modern reference carbon ($A_0 = 100\%$).

1. Calculate the central radiocarbon age t of the prehistoric cave charcoal ($t_{1/2} = 5730$ yr).
2. Using first-order differential error propagation, show that the uncertainty in age is given by $\Delta t = \dfrac{t_{1/2}}{\ln 2} \cdot \dfrac{\Delta A}{A}$.
3. Calculate the absolute uncertainty Δt and express the final dated age in standard scientific format: $t \pm \Delta t$ years BP (Before Present).

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.16

1. Central Age Calculation: Activity ratio: $\frac{A}{A_0} = 0.0185 \implies \frac{A_0}{A} = \frac{1}{0.0185} = 54.054$.

$$t = \frac{5730}{\ln 2} \ln(54.054) = 8266.64 \times 3.9900 = \mathbf{32\,984} \text{ years BP.}$$

2. Differential Error Propagation Derivation: Differentiating the dating equation $t = -\frac{1}{\lambda} \ln A + \text{const}$ with respect to A:

$$\frac{\mathrm{d}t}{\mathrm{d}A} = -\frac{1}{\lambda A} \implies |\Delta t| = \frac{1}{\lambda} \left| \frac{\Delta A}{A} \right| = \frac{\mathbf{t_{1/2}}}{\ln \mathbf{2}} \cdot \frac{\mathbf{\Delta A}}{\mathbf{A}}. \quad \text{Q.E.D.}$$

3. Uncertainty Calculation: Given $A = 1.85$ and $\Delta A = 0.05$:

$$\frac{\Delta A}{A} = \frac{0.05}{1.85} = 0.02703 \quad (2.70\%).$$

$$\Delta t = 8266.64 \times 0.02703 = \mathbf{223} \text{ years.}$$

Final calibrated age:

$$\mathbf{t = 33\,000 \pm 220} \text{ years BP} \quad \text{(Aurignacian paleolithic era).}$$

Exercise 3.17 · Uranium-Lead Geochronology of Ancient Zircons ★★★ Advanced

Zircon crystals ($ZrSiO_4$) incorporate uranium atoms into their crystal lattice during crystallization while rigorously rejecting lead (Pb^{2+}) due to ionic radius mismatch ($r(Zr^{4+}) = 84$ pm vs $r(Pb^{2+}) = 119$ pm). Thus, any ^{206}Pb present today arose exclusively from the alpha decay

cascade of Uranium-238 ($^{238}\text{U} \rightarrow {}^{206}\text{Pb}$, $t_{1/2} = 4.468 \times 10^9$ yr).

1. Derive the geochronological dating equation:
$$t = \frac{1}{\lambda} \ln\left(1 + \frac{N_{\text{Pb}}}{N_{\text{U}}}\right).$$
2. Mass spectrometric analysis of an ancient zircon crystal from the Jack Hills (Western Australia) yields an isotopic atom ratio of:
$$\frac{N(^{206}\text{Pb})}{N(^{238}\text{U})} = 0.9635.$$
Calculate the crystallization age t of this zircon crystal in billions of years (Ga).
3. Compare this age to the accepted scientific age of the Earth (4.54 Ga).

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.17

1. Derivation of Uranium-Lead Isochron Formula: Let N_0 be the initial number of uranium nuclei at crystallization ($t = 0$). Since no initial lead was incorporated ($N_{\text{Pb}}(0) = 0$), every radiogenic ^{206}Pb nucleus formed corresponds to one decayed ^{238}U nucleus:
$$N_{\text{Pb}}(t) = N_0 - N_{\text{U}}(t).$$
From exponential decay, $N_0 = N_{\text{U}}(t)e^{\lambda t}$. Substituting:
$$N_{\text{Pb}}(t) = N_{\text{U}}(t)e^{\lambda t} - N_{\text{U}}(t) = N_{\text{U}}(t)\left(e^{\lambda t} - 1\right).$$
Dividing by $N_{\text{U}}(t)$:
$$\frac{N_{\text{Pb}}}{N_{\text{U}}} = e^{\lambda t} - 1 \implies e^{\lambda t} = 1 + \frac{N_{\text{Pb}}}{N_{\text{U}}}.$$
Taking natural logarithms:
$$\mathbf{t} = \frac{\mathbf{1}}{\lambda} \ln\left(\mathbf{1} + \frac{\mathbf{N}_{\text{Pb}}}{\mathbf{N}_{\text{U}}}\right). \quad \text{Q.E.D.}$$
2. Age Calculation of Jack Hills Zircon: Decay constant of ^{238}U:
$$\lambda = \frac{\ln 2}{4.468 \times 10^9 \text{ yr}} = 1.55136 \times 10^{-10} \text{ yr}^{-1}.$$
Given ratio $N_{\text{Pb}}/N_{\text{U}} = 0.9635$:
$$1 + \frac{N_{\text{Pb}}}{N_{\text{U}}} = 1 + 0.9635 = 1.9635.$$

$$\ln(1.9635) = 0.67473.$$

$$t = \frac{0.67473}{1.55136 \times 10^{-10}\ \text{yr}^{-1}} = \mathbf{4.349 \times 10^9}\ \text{years} = \mathbf{4.35}\ \text{Ga}.$$

3. Geological Context: The Jack Hills zircon crystallized **4.35** billion years ago, merely 190 million years after the formation of the Earth (4.54 Ga). This confirms that solid continental crust and liquid water oceans existed astonishingly early in Earth's history!

Exercise 3.18 · **Branching Decay Kinetics and Potassium-Argon Dating** ★★★ Advanced

Potassium-40 ($^{40}_{19}\text{K}$) is an unstable primordial isotope that decays via two competing parallel pathways:

$$^{40}_{19}\text{K} + {}^{0}_{-1}\text{e} \longrightarrow {}^{40}_{18}\text{Ar} + \nu_e \quad \text{(Electron Capture, Branching Ratio } B_{\text{EC}} = 0.1072\text{)}$$
$$^{40}_{19}\text{K} \longrightarrow {}^{40}_{20}\text{Ca} + {}^{0}_{-1}\text{e} + \bar{\nu}_e \quad (\beta^- \text{ Decay, Branching Ratio } B_\beta = 0.8928)$$

The total half-life is $t_{1/2} = 1.248 \times 10^9$ years ($\lambda = \lambda_{\text{EC}} + \lambda_\beta$).

1. Derive the equation relating accumulated radiogenic Argon-40 to remaining Potassium-40:

$$N_{\text{Ar}}(t) = \frac{\lambda_{\text{EC}}}{\lambda} N_{\text{K}}(t) \left(e^{\lambda t} - 1\right).$$

2. A volcanic basalt sample erupted during an ancient rift event contains $m_{\text{K}} = 2.500\ \mu\text{g}$ of ^{40}K and $V_{\text{Ar}} = 1.250 \times 10^{-5}\ \text{cm}^3$ of radiogenic ^{40}Ar gas at STP ($V_0 = 22\,414\ \text{cm}^3 \cdot \text{mol}^{-1}$). Calculate the crystallization age of the volcanic rock.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.18

1. Derivation for Branching Decay: The total rate of loss of ^{40}K is $\frac{\text{d}N_{\text{K}}}{\text{d}t} = -\lambda N_{\text{K}}$, where $\lambda = \lambda_{\text{EC}} + \lambda_\beta$. The rate of generation of Argon-40 is exclusively through electron capture:

$$\frac{\text{d}N_{\text{Ar}}}{\text{d}t} = \lambda_{\text{EC}} N_{\text{K}}(t) = \lambda_{\text{EC}} N_0 e^{-\lambda t}.$$

Integrate from 0 to t (assuming initial molten rock outgassed all pre-existing argon, $N_{\text{Ar}}(0) = 0$):

$$N_{\text{Ar}}(t) = \lambda_{\text{EC}} N_0 \int_0^t e^{-\lambda t'}\,\text{d}t' = \frac{\lambda_{\text{EC}}}{\lambda} N_0 \left(1 - e^{-\lambda t}\right).$$

Substitute $N_0 = N_{\text{K}}(t) e^{\lambda t}$:

$$\mathbf{N_{Ar}(t) = \frac{\lambda_{EC}}{\lambda} N_K(t) \left(e^{\lambda t} - 1\right).} \quad \text{Q.E.D.}$$

2. Numerical Calculation: Total decay constant:

$$\lambda = \frac{\ln 2}{1.248 \times 10^9 \text{ yr}} = 5.554 \times 10^{-10} \text{ yr}^{-1}.$$

Branching fraction: $\frac{\lambda_{\text{EC}}}{\lambda} = 0.1072$. Number of ^{40}K nuclei:

$$N_{\text{K}} = \frac{2.500 \times 10^{-6} \text{ g}}{40.0 \text{ g} \cdot \text{mol}^{-1}} \times 6.022 \times 10^{23} = 3.7638 \times 10^{16} \text{ nuclei}.$$

Number of ^{40}Ar atoms:

$$N_{\text{Ar}} = \frac{1.250 \times 10^{-5} \text{ cm}^3}{22\,414 \text{ cm}^3 \cdot \text{mol}^{-1}} \times 6.022 \times 10^{23} = 3.3584 \times 10^{14} \text{ atoms}.$$

The ratio is:

$$\frac{N_{\text{Ar}}}{N_{\text{K}}} = \frac{3.3584 \times 10^{14}}{3.7638 \times 10^{16}} = 8.9229 \times 10^{-3}.$$

From the derived equation:

$$e^{\lambda t} - 1 = \frac{1}{0.1072} \times \left(8.9229 \times 10^{-3}\right) = 0.083236 \implies e^{\lambda t} = 1.083236.$$

$$\lambda t = \ln(1.083236) = 0.07995.$$

$$t = \frac{0.07995}{5.554 \times 10^{-10} \text{ yr}^{-1}} = \mathbf{1.440 \times 10^8} \text{ years} = \mathbf{144} \text{ Ma} \quad \text{(Late Jurassic period)}.$$

Exercise 3.19 · Cumulative Radiation Dose and Biological Half-Life ★★★ Advanced

In targeted thyroid radiotherapy, a patient receives an oral capsule containing $A_0 = 3.700$ GBq (100.0 mCi) of Iodine-131. The isotope is eliminated from the body simultaneously by radioactive decay (λ_p, physical half-life $t_p = 8.020$ days) and by biological excretion in urine and sweat (λ_b, biological clearance half-life $t_b = 21.00$ days).

1. Derive the **effective half-life** formula:
$$\frac{1}{t_{\text{eff}}} = \frac{1}{t_p} + \frac{1}{t_b} \implies t_{\text{eff}} = \frac{t_p \cdot t_b}{t_p + t_b}.$$
2. Calculate the effective half-life t_{eff} of Iodine-131 in this patient in days.
3. The total number of disintegrations occurring inside the patient's body over all time is $N_{\text{cum}} = \int_0^\infty A_{\text{in vivo}}(t)\mathrm{d}t$. Prove that $N_{\text{cum}} = \dfrac{A_0}{\lambda_{\text{eff}}}$, and compute its numerical value.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.19

1. Effective Half-Life Derivation: The in vivo disappearance of radioiodine follows two independent parallel first-order rate processes:

$$\frac{\mathrm{d}N}{\mathrm{d}t} = -(\lambda_p + \lambda_b)N(t) \implies \lambda_{\text{eff}} = \lambda_p + \lambda_b.$$

Expressing rate constants in terms of half-lives ($\lambda = \ln 2 / t_{1/2}$):

$$\frac{\ln 2}{t_{\text{eff}}} = \frac{\ln 2}{t_p} + \frac{\ln 2}{t_b} \implies \frac{\mathbf{1}}{\mathbf{t_{\text{eff}}}} = \frac{\mathbf{1}}{\mathbf{t_p}} + \frac{\mathbf{1}}{\mathbf{t_b}} \implies \mathbf{t_{\text{eff}}} = \frac{\mathbf{t_p \cdot t_b}}{\mathbf{t_p + t_b}}. \quad \text{Q.E.D.}$$

2. Numerical Calculation of t_{eff}: Given $t_p = 8.020$ days and $t_b = 21.00$ days:

$$t_{\text{eff}} = \frac{8.020 \times 21.00}{8.020 + 21.00} = \frac{168.42}{29.02} = \mathbf{5.804} \text{ days}.$$

3. Cumulative Total In Vivo Disintegrations: The in vivo activity is $A_{\text{in vivo}}(t) = A_0 e^{-\lambda_{\text{eff}} t}$. Integrating over $t \in [0, \infty)$:

$$N_{\text{cum}} = \int_0^\infty A_0 e^{-\lambda_{\text{eff}} t} \mathrm{d}t = A_0 \left[-\frac{e^{-\lambda_{\text{eff}} t}}{\lambda_{\text{eff}}} \right]_0^\infty = A_0 \left(0 - \left(-\frac{1}{\lambda_{\text{eff}}} \right) \right) = \frac{\mathbf{A_0}}{\lambda_{\text{eff}}}. \quad \text{Q.E.D.}$$

Convert t_{eff} to seconds:

$$t_{\text{eff}} = 5.804 \times 86\,400 \text{ s} = 5.0147 \times 10^5 \text{ s}.$$

$$\lambda_{\text{eff}} = \frac{\ln 2}{5.0147 \times 10^5 \text{ s}} = 1.3823 \times 10^{-6} \text{ s}^{-1}.$$

Given $A_0 = 3.700 \times 10^9$ Bq:

$$N_{\text{cum}} = \frac{3.700 \times 10^9 \text{ s}^{-1}}{1.3823 \times 10^{-6} \text{ s}^{-1}} = \mathbf{2.677 \times 10^{15}} \text{ nuclear disintegrations}.$$

Exercise 3.20 · The Bateman Differential Equations for Decay Cascades ★★★ Advanced

Consider a sequential radioactive decay chain with parent nucleus 1 decaying into daughter 2, which in turn decays into stable granddaughter 3:

$$1 \xrightarrow{\lambda_1} 2 \xrightarrow{\lambda_2} 3 \,(\text{stable}).$$

At time $t = 0$, only pure parent is present: $N_1(0) = N_0$ and $N_2(0) = 0$.

1. Write the coupled system of differential rate equations for $\mathrm{d}N_1/\mathrm{d}t$ and $\mathrm{d}N_2/\mathrm{d}t$.

2. Solve the system analytically using an integrating factor to prove the Bateman formula:

$$N_2(t) = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_0 \left(e^{-\lambda_1 t} - e^{-\lambda_2 t} \right).$$

3. Find the exact time $t_{\max}$ at which the daughter activity $A_2(t) = \lambda_2 N_2(t)$ reaches its maximum.
4. Apply this to the medical isotope generator system Molybdenum-99 ($\lambda_1 = 0.01050$ h^{-1}, $t_{1/2,1} = 66.0$ h) producing Technetium-99m ($\lambda_2 = 0.11533$ h^{-1}, $t_{1/2,2} = 6.01$ h). Calculate $t_{\max}$ in hours.

Graphic Tablet Workspace · 5 mm Grid for Stylus & Digital Pen Solution

Complete Step-by-Step Solution 3.20

1. Coupled Differential Equations:

$$\frac{\mathrm{d}N_1}{\mathrm{d}t} = -\lambda_1 N_1 \tag{3.1}$$

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = \lambda_1 N_1 - \lambda_2 N_2 \tag{3.2}$$

2. Analytical Solution by Integrating Factor: From (3.1), $N_1(t) = N_0 e^{-\lambda_1 t}$. Substitute into (3.2):

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} + \lambda_2 N_2 = \lambda_1 N_0 e^{-\lambda_1 t}.$$

Multiply by the integrating factor $e^{\lambda_2 t}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(N_2 e^{\lambda_2 t} \right) = \lambda_1 N_0 e^{(\lambda_2 - \lambda_1)t}.$$

Integrate both sides from 0 to t:

$$N_2(t) e^{\lambda_2 t} - 0 = \lambda_1 N_0 \left[\frac{e^{(\lambda_2 - \lambda_1)t} - 1}{\lambda_2 - \lambda_1} \right].$$

Multiplying across by $e^{-\lambda_2 t}$:

$$\mathbf{N_2(t) = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_0 \left(e^{-\lambda_1 t} - e^{-\lambda_2 t} \right).} \quad \text{Q.E.D.}$$

3. Determination of Maximum Time $t_{\max}$: Set the derivative $\frac{\mathrm{d}N_2}{\mathrm{d}t} = 0$:

$$-\lambda_1 e^{-\lambda_1 t_{\max}} + \lambda_2 e^{-\lambda_2 t_{\max}} = 0 \implies \lambda_1 e^{-\lambda_1 t_{\max}} = \lambda_2 e^{-\lambda_2 t_{\max}}.$$

$$\frac{e^{-\lambda_1 t_{\max}}}{e^{-\lambda_2 t_{\max}}} = \frac{\lambda_2}{\lambda_1} \implies e^{(\lambda_2 - \lambda_1) t_{\max}} = \frac{\lambda_2}{\lambda_1}.$$

Taking natural logarithms:

$$(\lambda_2 - \lambda_1)t_{\max} = \ln\left(\frac{\lambda_2}{\lambda_1}\right) \implies \mathbf{t}_{\max} = \frac{\ln(\boldsymbol{\lambda_2}/\boldsymbol{\lambda_1})}{\boldsymbol{\lambda_2} - \boldsymbol{\lambda_1}}.$$

4. Application to ^{99}Mo $\rightarrow$ ^{99m}Tc Generator: Given $\lambda_1 = 0.01050$ h^{-1} and $\lambda_2 = 0.11533$ h^{-1}:

$$\lambda_2 - \lambda_1 = 0.11533 - 0.01050 = 0.10483 \text{ h}^{-1}.$$

$$\frac{\lambda_2}{\lambda_1} = \frac{0.11533}{0.01050} = 10.9838 \implies \ln(10.9838) = 2.3964.$$

$$t_{\max} = \frac{2.3964}{0.10483 \text{ h}^{-1}} = \mathbf{22.86} \text{ hours} \approx \mathbf{23} \text{ hours}.$$

In hospital radiopharmacy practice, "milking" the molybdenum "cow" (technetium generator) is performed every 24 hours precisely because daughter technetium activity reaches its peak approximately 23 hours after each elution!